MULTISCALE MODEL. SIMUL.
Vol. 24, No. 3, pp. 1033–1059
© 2026 Society for Industrial and Applied Mathematics

Downloaded 07/11/26 to 202.113.11.53 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

AN EFFICIENT FOURIER SPECTRAL EIGENSOLVER FOR COMPUTING THE BOGOLIUBOV-DE GENNES EXCITATIONS OF SPIN-1 BOSE-EINSTEIN CONDENSATES*

YU LI†, ZHIXUAN LI‡, MANTING XIE§, AND YONG ZHANG¶

Abstract. In this paper, we propose a spectrally accurate solver for computing the Bogoliubov-de Gennes (BdG) excitations of spin-1 Bose-Einstein condensates (BECs), which are governed by the BdG equations, around the mean-field ground state. The BdG equations are essentially a constrained eigenvalue/eigenfunction system. First, we investigate their analytical properties, including exact eigenpairs, generalized nullspace, and bi-orthogonality of eigenspaces. Second, by combining the standard Fourier spectral method for spatial discretization and a stable Gram-Schmidt bi-orthogonal algorithm, we develop a subspace iterative solver for such a large-scale dense eigenvalue problem, and it proves to be numerically stable, efficient, and accurate. Our solver is matrix-free, and the operator-function evaluation is accelerated by a discrete fast Fourier transform (FFT) with almost optimal efficiency. Therefore, it is memory-friendly and efficient for large-scale problems. Furthermore, we give a rigorous and detailed numerical analysis of the stability and spectral convergence. Finally, we present extensive numerical results to illustrate the spectral accuracy and efficiency, and we investigate the excitation spectrum and Bogoliubov amplitudes around the ground state in 1–3 spatial dimensions.

Key words. eigenvalue problem, Bogoliubov-de Gennes excitations, Fourier spectral method, spin-1 Bose-Einstein condensates, bi-orthogonal structure, large-scale problem

MSC codes. 65M70, 68Q25, 65T50, 65N25, 65N35

DOI. 10.1137/25M1775853

1. Introduction.

Since the experiment realizations of degenerate spinor BECs with ultracold rubidium [15] and sodium [31], where the internal degrees of freedom of spin were activated by the optical dipole potential trap, there followed a vast number of physical experiments of various spinor BECs [26, 33] and theoretically investigations to explore various peculiar quantum phenomena [27]. In spinor BECs, particles of different hyperfine states allow for different angular momentum spaces, leading to very rich spin structures. Therefore, degenerate spinor quantum gases simultaneously exhibit magnetic and superfluid properties and shed light on a wide range of topics, such as topological quantum structures, fractional quantum Hall effects, etc. [21, 32].

At temperature T much smaller than the critical condensate temperature T_c [6], the spin-1 BEC is well described by the three-component wave function

*Received by the editors July 8, 2025; accepted for publication (in revised form) March 4, 2026; published electronically July 8, 2026.
https://doi.org/10.1137/25M1775853
Funding: The work of the first author was supported by National Natural Science Foundation of China grants 12301610 and 12271395. The work of the third author was supported by National Natural Science Foundation of China grants 12571437 and 12001402 and Tianjin University basic research fund 2025XJ21-0010. The work of the second and fourth authors was supported by National Key R&D Program of China grant 2024YFA1012803 and National Natural Science Foundation of China grant 12271400.

†Coordinated Innovation Center for Computable Modeling in Management Science, Tianjin University of Finance and Economics, Tianjin 300222, China (liyu@tjufe.edu.cn).
‡School of Mathematics, Sichuan University, Chengdu 610064, China (zhixuan_li@stu.scu.edu.cn).
§Corresponding author. School of Mathematics and KL-AAGDM, Tianjin University, Tianjin 300350, China (mtxie@tju.edu.cn).
¶Center for Applied Mathematics and KL-AAGDM, Tianjin University, Tianjin 300072, China, and State Key Laboratory of Synthetic Biology, Tianjin University, Tianjin 300072, China (Zhang_Yong@tju.edu.cn).

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.

Downloaded 07/11/26 to 202.113.11.53 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

$\Psi(\boldsymbol{x},t) = (\psi_1(\boldsymbol{x},t),\psi_0(\boldsymbol{x},t),\psi_{-1}(\boldsymbol{x},t))^\top$, whose evolution is governed by the coupled Gross–Pitaevskii equations (CGPEs) [16, 32, 38]. In dimensionless form, the d-dimensional ($d=1$, 2, or 3) CGPEs could be unified as [6, 7]

$$\mathrm{i}\partial_t\psi_1 = \left[-\frac{1}{2}\nabla^2 + V(\boldsymbol{x}) + \beta_n\rho + \beta_s\left(|\psi_0|^2+\rho_z\right)\right]\psi_1 + \beta_s\psi_0^2\bar{\psi}_{-1}, \tag{1.1a}$$

$$\mathrm{i}\partial_t\psi_0 = \left[-\frac{1}{2}\nabla^2 + V(\boldsymbol{x}) + \beta_n\rho + \beta_s\left(|\psi_1|^2+|\psi_{-1}|^2\right)\right]\psi_0 + 2\beta_s\psi_1\bar{\psi}_0\psi_{-1}, \tag{1.1b}$$

$$\mathrm{i}\partial_t\psi_{-1} = \left[-\frac{1}{2}\nabla^2 + V(\boldsymbol{x}) + \beta_n\rho + \beta_s\left(|\psi_0|^2-\rho_z\right)\right]\psi_{-1} + \beta_s\bar{\psi}_1\psi_0^2, \tag{1.1c}$$

where t denotes time and $\boldsymbol{x} = x \in \mathbb{R}$, $\boldsymbol{x} = (x,y)^\top \in \mathbb{R}^2$, and/or $\boldsymbol{x} = (x,y,z)^\top \in \mathbb{R}^3$ is the Cartesian coordinate vector, $\rho = \sum_{j=-1}^{1}|\psi_j|^2$ is the density function, and $\rho_z = |\psi_1|^2 - |\psi_{-1}|^2$. $V(\boldsymbol{x})$ is a real-valued external potential that is case-dependent, and one common choice is the following harmonic trapping potential:

$$V(\boldsymbol{x}) = \frac{1}{2}\begin{cases}\gamma_x^2x^2, & d=1,\\ \gamma_x^2x^2+\gamma_y^2y^2, & d=2,\\ \gamma_x^2x^2+\gamma_y^2y^2+\gamma_z^2z^2, & d=3.\end{cases} \tag{1.2}$$

Here, $\gamma_\alpha > 0$ ($\alpha = x,y,z$) are dimensionless constants proportional to the trapping frequency in the α-direction. Constant β_n denotes the mean-field interaction, and β_s represents spin-exchange interaction with positive/negative β_s corresponding to the antiferromagnetic/ferromagnetic case.

Introduce the triple of spin-1 Pauli matrices $\mathbf{S} = (\mathbf{S}_x,\mathbf{S}_y,\mathbf{S}_z)$ as

$$\mathbf{S}_x = \frac{1}{\sqrt{2}}\begin{bmatrix}0&1&0\\1&0&1\\0&1&0\end{bmatrix},\quad \mathbf{S}_y = \frac{\mathrm{i}}{\sqrt{2}}\begin{bmatrix}0&-1&0\\1&0&-1\\0&1&0\end{bmatrix},\quad \mathbf{S}_z = \begin{bmatrix}1&0&0\\0&0&0\\0&0&-1\end{bmatrix}$$

and the spin vector as

$$\mathbf{s}(\Psi) = (\mathbf{s}_x(\Psi),\mathbf{s}_y(\Psi),\mathbf{s}_z(\Psi))^\top = (\Psi^{\mathsf{H}}\mathbf{S}_x\Psi,\Psi^{\mathsf{H}}\mathbf{S}_y\Psi,\Psi^{\mathsf{H}}\mathbf{S}_z\Psi)^\top,$$

where $\bar{\boldsymbol{\zeta}}$, $\boldsymbol{\zeta}^{\mathsf{H}} := (\bar{\boldsymbol{\zeta}})^\top$ are the conjugate and conjugate transpose of vector $\boldsymbol{\zeta} \in \mathbb{C}^{3\times 1}$, respectively. Note that Pauli matrices are all Hermitian, i.e., $\mathbf{S}_j^{\mathsf{H}} = \mathbf{S}_j$ for $j = x,y,z$. The CGPEs (1.1) can be written in a compact form

$$\mathrm{i}\partial_t\Psi = \big(\big(L+\beta_n\rho\big)\mathbf{I}_3 + \beta_s\,\mathbf{S}\cdot\mathbf{s}(\Psi)\big)\,\Psi, \tag{1.3}$$

where $L = -\frac{1}{2}\nabla^2 + V(\boldsymbol{x})$, $\mathbf{I}_3$ is the identity matrix, and

$$\mathbf{S}\cdot\mathbf{s}(\Psi) := \mathbf{S}_x\,\mathbf{s}_x(\Psi) + \mathbf{S}_y\,\mathbf{s}_y(\Psi) + \mathbf{S}_z\,\mathbf{s}_z(\Psi).$$

The CGPEs (1.1) conserve three important quantities: the *mass*

$$\mathcal{N}(\Psi(\cdot,t)) = \|\Psi(\cdot,t)\|^2 := \int_{\mathbb{R}^d}\sum_{j=-1}^{1}|\psi_j(\boldsymbol{x})|^2\,\mathrm{d}\boldsymbol{x} \equiv \mathcal{N}(\Psi(\cdot,0)) = 1, \qquad t\ge 0,$$

the *magnetization* (with $-1 \le M \le 1$)

$$\mathcal{M}(\Psi(\cdot,t)) := \int_{\mathbb{R}^d}\left(|\psi_1(\boldsymbol{x})|^2 - |\psi_{-1}(\boldsymbol{x})|^2\right)\mathrm{d}\boldsymbol{x} \equiv \mathcal{M}(\Psi(\cdot,0)) = M, \qquad t\ge 0,$$

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.

Downloaded 07/11/26 to 202.113.11.53 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

and *energy*

$$\mathcal{E}(\Psi(\cdot,t)) := \int_{\mathbb{R}^d} \left\{ \sum_{j=-1}^{1} \Big(\frac{1}{2}|\nabla\psi_j(\boldsymbol{x})|^2 + V|\psi_j(\boldsymbol{x})|^2 \Big) + \frac{\beta_n}{2}|\Psi|^4 + \frac{\beta_s}{2}|\mathbf{s}(\Psi)|^2 \right\} \mathrm{d}\boldsymbol{x}$$
$$\equiv \mathcal{E}(\Psi(\cdot,0)), \quad t \geq 0,$$

where $|\Psi|$ is the length of vector Ψ, that is, $|\Psi| := \sqrt{|\psi_1|^2 + |\psi_0|^2 + |\psi_{-1}|^2} = \sqrt{\rho}$. The ground state, denoted by $\Phi_g = (\phi_1^g, \phi_0^g, \phi_{-1}^g)^\top$, is defined as the minimizer of the following problem:

$$\Phi_g = \arg\min_{\Phi \in \mathcal{S}_M} \mathcal{E}(\Phi), \tag{1.4}$$

where $\mathcal{S}_M$ is the constraint functional space

$$\mathcal{S}_M := \Big\{ \Phi = (\phi_1, \phi_0, \phi_{-1})^\top \;\Big|\; \|\Phi\|^2 = 1,\ \|\phi_1\|^2 - \|\phi_{-1}\|^2 = M,\ \mathcal{E}(\Phi) < \infty \Big\}.$$

The ground state Φ_g satisfies the following Euler–Lagrange equations:

$$\Lambda\Phi = \mathbf{H}\Phi := \Big(\big(L + \beta_n\rho\big)\mathbf{I}_3 + \beta_s\, \mathbf{S}\cdot\mathbf{s}(\Phi) \Big)\Phi, \tag{1.5}$$

where $\mathbf{H} = \mathrm{diag}(H_1, H_0, H_{-1})$ are the Hamiltonian operators, $\Phi = (\phi_1, \phi_0, \phi_{-1})^\top$, and $\Lambda = \mathrm{diag}(\mu_1, \mu_0, \mu_{-1})$, with entry μ_j, given below as the chemical potential associated with the jth component [6, 7],

$$\mu_j = \frac{\int_{\mathbb{R}^d} \overline{\phi_j} H_j \phi_j \mathrm{d}\boldsymbol{x}}{\int_{\mathbb{R}^d} |\phi_j|^2 \mathrm{d}\boldsymbol{x}}, \quad j = 1, 0, -1, \tag{1.6}$$

satisfying relation $\mu_1 + \mu_{-1} = 2\mu_0$. We can construct stationary states of (1.1) in the following way:

$$\Psi_s(\boldsymbol{x}, t) = \mathrm{diag}(e^{-\mathrm{i}\mu_1 t}, e^{-\mathrm{i}\mu_0 t}, e^{-\mathrm{i}\mu_{-1} t})\Phi_s(\boldsymbol{x}), \tag{1.7}$$

where $\Phi_s = (\phi_1^s, \phi_0^s, \phi_{-1}^s)^\top$ and $\{\mu_1, \mu_0, \mu_{-1}\}$ are solutions to the nonlinear eigenvalue problem (1.5).

On the mean-field level, the many-body effects are absent, and the CGPEs (1.1) prove to be valid for the spin-1 BECs [16, 32, 38]. However, due to the many-body nature of interatomic interactions, the system admits excitations even in its lowest-energy state. These excitations can be interpreted as quasi particles and are known as BdG excitations or, equivalently, elementary or collective excitations [8]. We first need to go beyond the mean-field theory to study BdG excitations. Under the proper assumption, the BdG excitations around the mean-field stationary state could be well described within the Bogoliubov theory, which resulted in the celebrated BdG equations [5, 12, 14, 18]. As the BdG excitations of BECs provide fundamental information about the ultracold quantum state, different phases of a trapped spin-1 BEC feature distinctive BdG excitations; therefore, they could be used to distinguish various phases.

The first experimental measurements of the lowest BdG modes of BECs were performed in [20]. Since then, great enthusiasm has been stimulated for mathematical and numerical investigation of BdG equations in the past few decades [17, 34]. Along the numerical front, there are many successful studies devoted to BdG corresponding

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.

Downloaded 07/11/26 to 202.113.11.53 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

to various BECs based on the well-known ARPACK library [22], and we refer the reader to [11, 12, 13, 35] for an incomplete list. The spatial discretization of the wavefunction, eigenfunction, and linear differential operators can be categorized into the finite difference method [13, 14], the finite-element method [11, 28], and the Fourier/sine spectral method [35, 39]. For a low dimension problem ($d = 1, 2$), the resulting eigensystem is solved with either a direct eigensolver, e.g., using "eigs" with MATLAB, or an iterative subspace solver, such as ARPACK or the locally optimal block preconditioned 4-d conjugate gradient [3, 4].

Up to now, there have been relatively few theoretical studies on the excitation of spin-1 BECs. Most numerical studies have focused on lower-dimensional cases ($d = 1$ or 2), for example the low-lying BdG excitations of quasi-two-dimensional rotating systems [12, 39] and cylindrically symmetric systems [37]. Chen et al. [10] directly solved the BdG equations for spin-orbit BECs using the Arnoldi method, while the Arnoldi method with ARPACK was adapted to study the dipolar spin-1 BEC [12]. Recently, Tang and coworkers [35, 39] constructed efficient and spectral numerical algorithms for the BdG equations of dipolar BECs in higher-dimensional spaces.

As far as we know, there have been quite a few mathematical or numerical studies on the BdG excitations of spin-1 BECs. Hence, it is necessary to develop mathematical theories and construct an accurate and efficient eigensolver. To numerically study the BdG excitations, there are two main challenges: (i) an accurate stationary state solver, especially the ground state solver; (ii) an accurate and efficient BdG solver. As for the computation of the ground state, various numerical methods have been proposed for the spin-1 BEC, such as the gradient flow method [6, 7], the gradient flow with the Lagrange multiplier (GFLM) method [24], the regularized Newton method [36], the preconditioned conjugate gradient (PCG) method [1], etc. Here we adopt the GFLM to compute the ground state due to its simplicity and effectiveness.

As is known, the structure of the eigenvalue distribution and the generalized nullspace is of essential importance to eigensolver performance in terms of efficiency, accuracy, and stability. It is noted that eigenvalues, coming in negative/positive pairs, are real and span over the real line to infinity, and the eigenspaces corresponding to eigenvalues of different magnitudes are bi-orthogonal [23]. More importantly, for eigenvalue zero, the algebraic multiplicity may not be equal to its geometric multiplicity, thus leading to a rich and complex generalized nullspace structure. However, such peculiar inherent structure and analytical properties have not been considered or taken into account in the state-of-the-art popular solvers; for example, the ARPACK library is much more of a general-purpose eigensolver. Direct application of the aforementioned solvers may encounter slow convergence or even divergence, especially at eigenvalue zero, and the performance will definitely deteriorate in high-dimensional problems.

To make it more challenging, the eigenfunctions are usually discretized by the Fourier spectral method; therefore, the resulting discrete eigensystem is a fully-populated dense matrix, which renders prohibitively huge memory costs if one chooses to store the matrix explicitly. Therefore, it is imperative to adopt some iterative algorithm, where one only needs to provide a matrix-vector product or operator-function evaluation, so as to bypass the huge explicit matrix storage and gain good parallel scalability for large-scale high-dimensional problems.

Overall, the main contributions of this paper are fourfold:

(1) derive the BdG equations for spin-1 BECs around the ground state and investigate its mathematical structures, including analytical eigenvalues/eigenfunctions, generalized nullspace, and the bi-orthogonality between eigenspaces;

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.

(2) develop a bi-orthogonal structure-preserving Fourier spectral eigensolver, and propose an efficient implementation of the matrix-vector product based on the Fourier spectral method;
(3) prove the numerical stability and spectral convergence;
(4) verify its spectral accuracy, efficiency, and scalability, and numerically study the excitation spectrum and Bogoliubov amplitudes around the ground state with different parameters in 1D, 2D, and 3D.

The rest of the paper is organized as follows: In section 2, we introduce the BdG equations and derive some analytical properties. In section 3, we present details of the Fourier spectral method for space discretization and propose an efficient eigensolver. Extensive numerical examples are shown in section 4 to confirm the performance of our method, together with some applications to study the solutions to the BdG equations with different parameters in 1D, 2D, and 3D. Finally, conclusions are drawn in section 5.

2. The BdG equations and their properties.

2.1. The Bogoliubov–de Gennes equations. To characterize the BdG excitations of a spin-1 BEC, the Bogoliubov theory [5, 12] begins with the ground state Φ_g of the CGPEs (1.3), which are also the solution of the nonlinear eigenvalue problems (1.5) with corresponding chemical potentials μ_1, μ_0, and μ_{-1}, and assumes the evolution of CGPEs (1.3) is around Φ_g. The corresponding wave function Ψ takes the following form [19, 35]:

$$\Psi(\boldsymbol{x},t)=\operatorname{diag}\big(e^{-\mathrm{i}\mu_1 t},e^{-\mathrm{i}\mu_0 t},e^{-\mathrm{i}\mu_{-1} t}\big)\left[\Phi_g(\boldsymbol{x})+\varepsilon\sum_{\ell=1}^{\infty}\Big(\boldsymbol{u}^\ell(\boldsymbol{x})e^{-\mathrm{i}\omega_\ell t}+\bar{\boldsymbol{v}}^\ell(\boldsymbol{x})e^{\mathrm{i}\omega_\ell t}\Big)\right] \tag{2.1}$$

for $\boldsymbol{x}\in\mathbb{R}^d$ and $t>0$. Here, $0<\varepsilon\ll 1$ is a small quantity used to control the population of quasi-particle excitation, $\omega_\ell\in\mathbb{C}$ is the frequency of the excitations to be determined, and $\boldsymbol{u}^\ell,\boldsymbol{v}^\ell\in H^1(\mathbb{R}^3;\mathbb{C}^3)$ are the Bogoliubov excitation modes satisfying the normalization condition

$$\int_{\mathbb{R}^d}\big(|\boldsymbol{u}^\ell(\boldsymbol{x})|^2-|\boldsymbol{v}^\ell(\boldsymbol{x})|^2\big)\,\mathrm{d}\boldsymbol{x}=\int_{\mathbb{R}^d}\sum_{j=-1}^{1}\big(|u_j^\ell(\boldsymbol{x})|^2-|v_j^\ell(\boldsymbol{x})|^2\big)\,\mathrm{d}\boldsymbol{x}=1,\quad \ell\in\mathbb{Z}^+, \tag{2.2}$$

where $\boldsymbol{u}^\ell=(u_1^\ell,u_0^\ell,u_{-1}^\ell)^\top,\boldsymbol{v}^\ell=(v_1^\ell,v_0^\ell,v_{-1}^\ell)^\top$. Plugging (2.1) into (1.3), by collecting linear terms in ε and separating frequency $e^{-i\omega_\ell t}$, we obtain the BdG equations as follows:

$$\begin{bmatrix}\mathcal{A} & \mathcal{B}\\ -\mathcal{C} & -\mathcal{D}\end{bmatrix}\begin{bmatrix}\boldsymbol{u}\\ \boldsymbol{v}\end{bmatrix}=\omega\begin{bmatrix}\boldsymbol{u}\\ \boldsymbol{v}\end{bmatrix}, \tag{2.3}$$

with constraint

$$\int_{\mathbb{R}^d}\big(|\boldsymbol{u}(\boldsymbol{x})|^2-|\boldsymbol{v}(\boldsymbol{x})|^2\big)\,\mathrm{d}\boldsymbol{x}=1, \tag{2.4}$$

where all the subscripts ℓ are omitted hereafter for simplicity, ω is the excitation energy, and the operators are given explicitly:

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.

Downloaded 07/11/26 to 202.113.11.53 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

$$\mathcal{A} = (L\mathbf{I}_3 - \Lambda) + \beta_n\big(\rho\mathbf{I}_3 + \Phi_g\Phi_g^{\mathsf{H}}\big) + \beta_s\bigg(\mathbf{S}\cdot\mathbf{s}(\Phi_g) + \sum_{j=x,y,z}\mathbf{S}_j\Phi_g\Phi_g^{\mathsf{H}}\mathbf{S}_j^{\mathsf{H}}\bigg), \tag{2.5}$$

$$\mathcal{B} = \beta_n\Phi_g\Phi_g^{\top} + \beta_s\sum_{j=x,y,z}\mathbf{S}_j\Phi_g\Phi_g^{\top}\mathbf{S}_j^{\top}, \tag{2.6}$$

$$\mathcal{C} = \beta_n\bar{\Phi}_g\Phi_g^{\mathsf{H}} + \beta_s\sum_{j=x,y,z}\bar{\mathbf{S}}_j\bar{\Phi}_g\Phi_g^{\mathsf{H}}\mathbf{S}_j^{\mathsf{H}}, \tag{2.7}$$

$$\mathcal{D} = (L\mathbf{I}_3 - \Lambda) + \beta_n\big(\rho\mathbf{I}_3 + \bar{\Phi}_g\Phi_g^{\top}\big) + \beta_s\bigg(\bar{\mathbf{S}}\cdot\mathbf{s}(\Phi_g) + \sum_{j=x,y,z}\bar{\mathbf{S}}_j\bar{\Phi}_g\Phi_g^{\top}\mathbf{S}_j^{\top}\bigg). \tag{2.8}$$

It is easy to check that $\mathcal{A}$ and $\mathcal{D}$ are both Hermitian operators, i.e., $\mathcal{A}^* = \mathcal{A}$, $\mathcal{D}^* = \mathcal{D}$, and $\mathcal{B}^* = \mathcal{C}$, where symbol $*$ denotes the operator adjoint associated with inner product $\langle \boldsymbol{f}, \boldsymbol{g}\rangle := \sum_{j=-1}^{1}\int_{\mathbb{R}^d} f_j(\boldsymbol{x})\overline{g_j(\boldsymbol{x})}\,\mathrm{d}\boldsymbol{x}$. Furthermore, we have $\overline{\mathcal{A}\boldsymbol{g}} = \mathcal{D}\bar{\boldsymbol{g}}$, $\overline{\mathcal{B}\boldsymbol{g}} = \mathcal{C}\bar{\boldsymbol{g}}$ $\forall\,\boldsymbol{g}\in H^1(\mathbb{R}^d;\mathbb{C}^3)$, and it immediately implies that the operators' finite-dimensional subspace representations, denoted by matrices A, B, C, and D, are either Hermitian or symmetric, that is,

$$A^{\mathsf{H}} = A, \quad D = \overline{A}, \quad B^{\top} = B, \quad \text{and} \quad C = \overline{B}. \tag{2.9}$$

The matrices representation of (2.3) reads as follows:

$$\begin{bmatrix} A & B \\ -\overline{B} & -\overline{A}\end{bmatrix}\begin{bmatrix}\mathbf{u}\\ \mathbf{v}\end{bmatrix} = \omega\begin{bmatrix}\mathbf{u}\\ \mathbf{v}\end{bmatrix}, \tag{2.10}$$

and it coincides with the Bethe–Salpeter Hamiltonian (BSH) matrix arising from optical absorption spectrum analysis [29].

2.2. Analytical properties. In this section, we derive some analytical properties of the BdG equations as well as the structure of the generalized nullspace for both the ferromagnetic and the antiferromagnetic cases. They might serve as benchmarks for numerical solutions or help to design an efficient eigensolver.

THEOREM 2.1 (symmetric distribution). *If* $\{\omega; \boldsymbol{u}, \boldsymbol{v}\}$ *is a solution pair to the BdG equations* (2.3)*, then* $\{-\bar{\omega}; \bar{\boldsymbol{v}}, \bar{\boldsymbol{u}}\}$ *is also a solution pair. Furthermore, If* $(\boldsymbol{u}, \boldsymbol{v})$ *satisfies the normalization constraint* (2.4)*, i.e., the BdG excitations, the frequency* ω *is real.*

Proof. The first conclusion can be proved by taking the conjugate of (2.3). Multiplying the first/second equation of (2.3) by $\boldsymbol{u}/\boldsymbol{v}$, respectively, and integrating each equation concerning $\boldsymbol{x}$, we subtract the second integration from the first one to obtain the following:

$$\langle \boldsymbol{u}, \mathcal{A}\boldsymbol{u}\rangle + \langle \boldsymbol{u}, \mathcal{B}\boldsymbol{v}\rangle + \langle \boldsymbol{v}, \mathcal{C}\boldsymbol{u}\rangle + \langle \boldsymbol{v}, \mathcal{D}\boldsymbol{v}\rangle = \bar{\omega}\int_{\mathbb{R}^d}(|\boldsymbol{u}(\boldsymbol{x})|^2 - |\boldsymbol{v}(\boldsymbol{x})|^2)\,\mathrm{d}\boldsymbol{x}. \tag{2.11}$$

From (2.4), we can see that ω is real. ∎

As shown in [6], under appropriate assumptions, the ground states can be chosen as real-valued functions. Therefore, in this article, we shall focus on the real ground state hereafter, and all operators involved are real. To be more specific,

$$\mathcal{D} = \mathcal{A}, \quad \mathcal{C} = \mathcal{B}.$$

The BdG equation (2.3) is a linear response eigenvalue problem of the following form:

$$\begin{bmatrix} \mathcal{A} & \mathcal{B} \\ -\mathcal{B} & -\mathcal{A}\end{bmatrix}\begin{bmatrix}\boldsymbol{u}\\ \boldsymbol{v}\end{bmatrix} = \omega\begin{bmatrix}\boldsymbol{u}\\ \boldsymbol{v}\end{bmatrix}. \tag{2.12}$$

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.

Downloaded 07/11/26 to 202.113.11.53 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

Downloaded 07/11/26 to 202.113.11.53 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

By applying a change of variables

$$\boldsymbol{u} = \boldsymbol{f} + \boldsymbol{g},\ \boldsymbol{v} = \boldsymbol{f} - \boldsymbol{g}, \tag{2.13}$$

the above equation can be reformulated as

$$\mathcal{H}\begin{bmatrix}\boldsymbol{f}\\ \boldsymbol{g}\end{bmatrix} := \begin{bmatrix}\mathcal{O} & \mathcal{H}_-\\ \mathcal{H}_+ & \mathcal{O}\end{bmatrix}\begin{bmatrix}\boldsymbol{f}\\ \boldsymbol{g}\end{bmatrix} = \omega\begin{bmatrix}\boldsymbol{f}\\ \boldsymbol{g}\end{bmatrix}, \tag{2.14}$$

where $\mathcal{H}_- := \mathcal{A}-\mathcal{B}$ and $\mathcal{H}_+ := \mathcal{A}+\mathcal{B}$ are both Hermitian operators, and the constraint (2.4) is reformulated in $(\boldsymbol{f},\boldsymbol{g})$ as

$$\langle \boldsymbol{f}, \boldsymbol{g}\rangle = \frac{1}{4}. \tag{2.15}$$

Remark 2.2. If $\mathcal{H}_+$ is positive definite, according to Lemma 2.3 of [23], the constraint (2.4) implies that the eigenvalue ω is nonzero; therefore, we shall only focus on the *nonzero* eigenvalues and their eigenfunctions.

THEOREM 2.3 (analytical eigenpairs). *Let* $\Phi_g = (\phi_1^g, \phi_0^g, \phi_{-1}^g)^\top$ *be real-valued ground states for* (1.1) *with harmonic trapping potential* (1.2)*, and we have analytical solutions to the BdG equation* (2.3) *as follows:*

$$\begin{aligned}&\{\omega_\alpha; \boldsymbol{u}_\alpha, \boldsymbol{v}_\alpha\}\\ &:= \left\{\gamma_\alpha; \frac{1}{\sqrt{2}}\left(\gamma_\alpha^{-\frac{1}{2}}\partial_\alpha\Phi_g - \gamma_\alpha^{\frac{1}{2}}\alpha\Phi_g\right), \frac{1}{\sqrt{2}}\left(\gamma_\alpha^{-\frac{1}{2}}\partial_\alpha\Phi_g + \gamma_\alpha^{\frac{1}{2}}\alpha\Phi_g\right)\right\},\end{aligned} \tag{2.16}$$

with $\alpha = x$ *in one dimension,* $\alpha = x, y$ *in two dimensions, and* $\alpha = x, y, z$ *in three dimensions.*

Proof. For simplicity, we only prove the $\alpha = x$ case, and extensions to other spatial variables are similar. Differentiating (1.5) with respect to x and combining the $\mathcal{H}_+$ definition, we derive

$$\mathcal{H}_+(\partial_x\Phi_g) = \gamma_x(-\gamma_x x\Phi_g),$$

with $\partial_x\Phi_g = (\partial_x\phi_1^g, \partial_x\phi_0^g, \partial_x\phi_{-1}^g)^\top$. Applying $\mathcal{H}_-$ on $-\gamma_x x\Phi_g$, we have

$$\mathcal{H}_-(-\gamma_x x\Phi_g) = \gamma_x(\partial_x\Phi_g).$$

Therefore, we can see that $(\partial_x\Phi_g^\top, -\gamma_x x\Phi_g^\top)^\top$ solves (2.14) with $\omega = \gamma_x$. With simple calculations, we have

$$\langle -\gamma_x x\Phi_g, \partial_x\Phi_g\rangle = \int_{\mathbb{R}^2}\sum_{j=-1}^{1}\left(-\gamma_x x\phi_j^g\partial_x\phi_j^g\right)\mathrm{d}x\mathrm{d}y = \frac{\gamma_x}{2}.$$

Therefore, the normalized function $(\boldsymbol{f},\boldsymbol{g}) = (\frac{1}{\sqrt{2}}\gamma_x^{-\frac{1}{2}}\partial_x\Phi_g, -\frac{1}{\sqrt{2}}\gamma_x^{\frac{1}{2}}x\Phi_g)$ solves (2.14) with eigenvalue $\omega = \gamma_x$. Finally, we derive the following analytic solutions:

$$\omega_x = \gamma_x, \quad \boldsymbol{u}_x = \frac{1}{\sqrt{2}}\left(\gamma_x^{-\frac{1}{2}}\partial_x\Phi_g - \gamma_x^{\frac{1}{2}}x\Phi_g\right), \quad \boldsymbol{v}_x = \frac{1}{\sqrt{2}}\left(\gamma_x^{-\frac{1}{2}}\partial_x\Phi_g + \gamma_x^{\frac{1}{2}}x\Phi_g\right). \qquad \square$$

THEOREM 2.4 (bi-orthogonality). *Assuming* $\{\omega_i; \boldsymbol{f}_i, \boldsymbol{g}_i\}_{i=1}^2$ *are eigenpairs of* (2.14) *with eigenvalues of different magnitudes, i.e.,* $|\omega_1| \neq |\omega_2|$*, the following bi-orthogonal properties hold true:*

$$\langle \boldsymbol{f}_1, \boldsymbol{g}_2\rangle = \langle \boldsymbol{f}_2, \boldsymbol{g}_1\rangle = 0.$$

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.

Downloaded 07/11/26 to 202.113.11.53 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

Proof. Using (2.14), we have

$$\omega_i^2 \langle \boldsymbol{f}_i, \boldsymbol{g}_j \rangle = \langle \mathcal{H}_- \mathcal{H}_+ \boldsymbol{f}_i, \boldsymbol{g}_j \rangle = \langle \boldsymbol{f}_i, \mathcal{H}_+ \mathcal{H}_- \boldsymbol{g}_j \rangle = \omega_j^2 \langle \boldsymbol{f}_i, \boldsymbol{g}_j \rangle,$$

which means

$$(\omega_i^2 - \omega_j^2) \langle \boldsymbol{f}_i, \boldsymbol{g}_j \rangle = 0.$$

Since $|\omega_i| \neq |\omega_j|$, we obtain $\langle \boldsymbol{f}_i, \boldsymbol{g}_j \rangle = 0$ for $i \neq j$. ∎

2.3. Generalized nullspace. As is known, the algebraic/geometric multiplicity of eigenvalue zero and the structure of its associated generalized nullspace are of great importance to the eigensolver's performance in terms of convergence, accuracy, and efficiency. The generalized nullspace of $\mathcal{H}$ is defined as the nullspace of $\mathcal{H}^p$ for some positive integer p such that

$$\mathrm{null}(\mathcal{H}^p) = \mathrm{null}(\mathcal{H}^{p+1}).$$

For compact operator $\mathcal{H}$, the integer p is finite and $\mathrm{null}(\mathcal{H}^p) = \mathrm{null}(\mathcal{H}^q)$ holds true for any greater integer q, i.e, $q \geq p$. In this subsection, we shall elaborate on the structure of the generalized nullspace for ferromagnetic and antiferromagnetic systems, which is, of course, closely connected with the nullspace of $\mathcal{H}_-$ and $\mathcal{H}_+$.

Case I: Ferromagnetic system ($\beta_s < 0$). For the ferromagnetic system, when the following conditions hold, i.e.,

$$M \in (-1, 1),\ \beta_s < 0,\ \text{and}\ \beta_n + \beta_s \geq 0, \tag{2.17}$$

we have the single-mode approximation; i.e., each component of the ground state Φ_g is identical up to a constant factor, which is given in Lemma 2.5.

LEMMA 2.5 (single-mode approximation (SMA) [6, Theorem 4.2]). *Under conditions* (2.17)*, there exists a ground state solution* $\Phi_g = \boldsymbol{a}\,\phi_g$, $\boldsymbol{a} = \big(\frac{1+M}{2}, (\frac{1-M^2}{2})^{1/2}, \frac{1-M}{2}\big)^\top$*, and the chemical potentials are identical, i.e.,* $\mu_1 = \mu_0 = \mu_{-1}$*. Here,* ϕ_g *is the unique positive minimizer of the following energy functional:*

$$E_{SMA}(\phi) = \int_{\mathbb{R}^d} \left[\frac{1}{2} |\nabla \phi|^2 + V(\boldsymbol{x}) |\phi|^2 + \frac{\beta_n + \beta_s}{2} |\phi|^4 \right] \mathrm{d}\boldsymbol{x}$$

under constraint $\mathcal{S} := \{\phi \in L^2(\mathbb{R}^d) \big| \|\phi\|^2 = 1, E_{SMA}(\phi) < \infty\}$.

THEOREM 2.6 (nullspace of $\boldsymbol{\mathcal{H}_-}$ and $\boldsymbol{\mathcal{H}_+}$ for ferromagnetic system). *Under conditions* (2.17)*, we derive the nullspace of* $\mathcal{H}_-$ *and* $\mathcal{H}_+$ *as follows:*

$$\mathrm{null}(\mathcal{H}_-) = \mathrm{span}\{\Phi_1, \Phi_2\}, \qquad \mathrm{null}(\mathcal{H}_+) = \mathrm{span}\{\Phi_2\}, \tag{2.18}$$

where $\Phi_1 = \Phi_g = \boldsymbol{a}\,\phi_g$, $\Phi_2 = \boldsymbol{b}\,\phi_g$ *with* $\boldsymbol{b} = \big((\frac{1-M^2}{2})^{1/2}, -M, -(\frac{1-M^2}{2})^{1/2}\big)^\top$.

Moreover, $\mathcal{H}_-$ and $\mathcal{H}_+$ are positive semidefinite.

Proof. It is straightforward to check that

$$\begin{aligned} \mathcal{H}_- = \mathcal{A} - \mathcal{B} = &\Big(L + (\beta_n + \beta_s)\phi_g^2 - \mu\Big)\mathbf{I}_3 \\ &+ \beta_s \phi_g^2 \Big(\mathbf{S} \cdot \mathbf{s}(\boldsymbol{a}) + \mathbf{S}_y \boldsymbol{a}\boldsymbol{a}^\top \mathbf{S}_y^{\mathsf{H}} - \mathbf{S}_y \boldsymbol{a}\boldsymbol{a}^\top \mathbf{S}_y^\top - \mathbf{I}_3\Big), \end{aligned}$$

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.

Downloaded 07/11/26 to 202.113.11.53 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

where $\mu = \mu_1 = \mu_0 = \mu_{-1}$ in Lemma 2.5. Moreover, we have

$$\mathcal{W}^{-1}\Big(\mathbf{S}\cdot\mathbf{s}(\boldsymbol{a}) + \mathbf{S}_y\boldsymbol{a}\boldsymbol{a}^\top\mathbf{S}_y^{\mathtt{H}} - \mathbf{S}_y\boldsymbol{a}\boldsymbol{a}^\top\mathbf{S}_y^\top - \mathbf{I}_3\Big)\mathcal{W} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -2 \end{bmatrix},$$

where $\mathcal{W} = [\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}]$ is an orthogonal symmetric matrix with the third column $\boldsymbol{c} = \big(\frac{1-M}{2}, -(\frac{1-M^2}{2})^{1/2}, \frac{1+M}{2}\big)^\top$. Then we can diagonalize operator $\mathcal{H}_-$ using $\mathcal{W}$ as $\mathcal{W}^{-1}\mathcal{H}_-\mathcal{W} = \mathcal{K}$, where $\mathcal{K}$ reads explicitly as follows

$$\mathcal{K} = \operatorname{diag}\Big(L + (\beta_n + \beta_s)\phi_g^2 - \mu, L + (\beta_n + \beta_s)\phi_g^2 - \mu, L + (\beta_n - \beta_s)\phi_g^2 - \mu\Big).$$

According to Lemmas 1 and 2 in [9], $\operatorname{null}\big(L + (\beta_n + \beta_s)\phi_g^2 - \mu\big) = \operatorname{span}\{\phi_g\}$, and $L + (\beta_n - \beta_s)\phi_g^2 - \mu$ is symmetric positive definite. Hence, we can derive the nullspace of $\mathcal{K}$ as

$$\operatorname{null}(\mathcal{K}) = \operatorname{span}\left\{\begin{bmatrix}1\\0\\0\end{bmatrix}\phi_g,\ \begin{bmatrix}0\\1\\0\end{bmatrix}\phi_g\right\}.$$

Therefore, we obtain

$$\operatorname{null}(\mathcal{H}_-) = \operatorname{span}\{\Phi_1, \Phi_2\} = \operatorname{span}\{\boldsymbol{a}\,\phi_g, \boldsymbol{b}\,\phi_g\}.$$

Similarly, we have

$$\operatorname{null}(\mathcal{H}_+) = \operatorname{span}\{\Phi_2\}.$$

The proofs are similar, and we choose to omit details for brevity. ∎

THEOREM 2.7 (generalized nullspace of $\boldsymbol{\mathcal{H}}$ for ferromagnetic system). *Under conditions* (2.17)*, we have the nullspace of* $\mathcal{H}$ *as*

$$\operatorname{null}(\mathcal{H}) = \operatorname{span}\left\{\begin{bmatrix}\mathbf{0}\\\Phi_1\end{bmatrix}, \begin{bmatrix}\mathbf{0}\\\Phi_2\end{bmatrix}, \begin{bmatrix}\Phi_2\\\mathbf{0}\end{bmatrix}\right\} \tag{2.19}$$

and the generalized nullspace as

$$\operatorname{null}(\mathcal{H}^3) = \operatorname{null}(\mathcal{H}^2) = \operatorname{null}(\mathcal{H}) \oplus \operatorname{span}\left\{\begin{bmatrix}\widehat{\Phi}_1\\\mathbf{0}\end{bmatrix}\right\}, \tag{2.20}$$

where $\widehat{\Phi}_1$*, satisfying* $\mathcal{H}_+\widehat{\Phi}_1 = \Phi_1$*, is independent of* Φ_2*.*

Proof. For any $(\boldsymbol{f}^\top, \boldsymbol{g}^\top)^\top \in \operatorname{null}(\mathcal{H})$, we have $\boldsymbol{f} \in \operatorname{null}(\mathcal{H}_+)$ and $\boldsymbol{g} \in \operatorname{null}(\mathcal{H}_-)$. According to Theorem 2.6, we can prove (2.19). Next, we shall investigate the nullspace of $\mathcal{H}^2$. For any $(\boldsymbol{f}^\top, \boldsymbol{g}^\top)^\top \in \operatorname{null}(\mathcal{H}^2)$, we have

$$\begin{bmatrix}\mathcal{O} & \mathcal{H}_-\\ \mathcal{H}_+ & \mathcal{O}\end{bmatrix}\begin{bmatrix}\boldsymbol{f}\\\boldsymbol{g}\end{bmatrix} \in \operatorname{null}(\mathcal{H}). \tag{2.21}$$

It is worthy to note that Φ_2 does not belong to range of $\mathcal{H}_+$, i.e., $\Phi_2 \notin \mathcal{R}(\mathcal{H}_+)$. Otherwise, there exists a solution to $\mathcal{H}_+\boldsymbol{f} = \Phi_2$, and then we derive a contradiction:

$$0 < \langle\Phi_2, \Phi_2\rangle = \langle\Phi_2, \mathcal{H}_+\boldsymbol{f}\rangle = \langle\mathcal{H}_+\Phi_2, \boldsymbol{f}\rangle = 0.$$

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.

Downloaded 07/11/26 to 202.113.11.53 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

Similarly, we can prove that $\Phi_2 \notin \mathcal{R}(\mathcal{H}_-)$. Therefore, (2.21) is equivalent to

$$\mathcal{H}_- \boldsymbol{g} = \mathbf{0}, \quad \mathcal{H}_+ \boldsymbol{f} = \Phi_1.$$

Since $\mathcal{H}_+$ is a self-adjoint compact operator, using the spectral theory of operators, the fact that $\Phi_1 \notin \mathrm{null}(\mathcal{H}_+)$ leads to $\Phi_1 \in \mathcal{R}(\mathcal{H}_+)$. That is, there exists a function $\widehat{\Phi}_1$ such that $\mathcal{H}_+ \widehat{\Phi}_1 = \Phi_1$. Then we have

$$\mathrm{null}(\mathcal{H}^2) = \mathrm{null}(\mathcal{H}) \oplus \mathrm{span}\left\{ \begin{bmatrix} \widehat{\Phi}_1 \\ \mathbf{0} \end{bmatrix} \right\}.$$

Finally, we study the nullspace of $\mathcal{H}^3$. Similarly, if $\mathrm{null}(\mathcal{H}^2) \subsetneq \mathrm{null}(\mathcal{H}^3)$, we have the following equivalence:

$$\begin{bmatrix} \mathcal{O} & \mathcal{H}_- \\ \mathcal{H}_+ & \mathcal{O} \end{bmatrix} \begin{bmatrix} \boldsymbol{f} \\ \boldsymbol{g} \end{bmatrix} = \begin{bmatrix} \widehat{\Phi}_1 \\ \mathbf{0} \end{bmatrix} \Longleftrightarrow \begin{cases} \mathcal{H}_- \boldsymbol{g} = \widehat{\Phi}_1, \\ \mathcal{H}_+ \boldsymbol{f} = \mathbf{0}. \end{cases}$$

However, the above equation does not admit any solutions, and it can be proved using the following argument. If there exists a solution $\boldsymbol{g}$ satisfying $\mathcal{H}_- \boldsymbol{g} = \widehat{\Phi}_1$, then we have the following contradiction:

$$0 = \langle \mathcal{H}_- \Phi_1, \boldsymbol{g} \rangle = \langle \Phi_1, \mathcal{H}_- \boldsymbol{g} \rangle = \langle \Phi_1, \widehat{\Phi}_1 \rangle = \langle \mathcal{H}_+ \widehat{\Phi}_1, \widehat{\Phi}_1 \rangle > 0.$$

That is to say, $\mathrm{null}(\mathcal{H}^3) = \mathrm{null}(\mathcal{H}^2)$, and the proof is completed. ∎

Case II: Antiferromagnetic system ($\beta_s > 0$). For the anti-ferromagnetic system, when the following conditions hold, i.e.,

$$M \in (-1, 1), \ \beta_s > 0, \ \text{and} \ \beta_n \geq 0, \tag{2.22}$$

we have the vanishing phenomenon; i.e., the ground state $\Phi_g = (\phi_1^g, \phi_0^g, \phi_{-1}^g)^\top$ satisfies $\phi_0^g = 0$, which is presented in Lemma 2.8.

LEMMA 2.8 ([6, Theorem 4.3]). *Under conditions* (2.22)*, there exists a ground state solution* $\Phi_g = (\phi_1^g, 0, \phi_{-1}^g)^\top$ *and* (ϕ_1^g, ϕ_{-1}^g) *is a minimizer of the energy functional*

$$E_0(\phi_1, \phi_{-1}) = \int_{\mathbb{R}^d} \left[\sum_{j=\pm 1} \left(\frac{|\nabla \phi_j|^2}{2} + V|\phi_j|^2 \right) + \frac{\beta_n + \beta_s}{2} (|\phi_1|^4 + |\phi_{-1}|^4) + (\beta_n - \beta_s)|\phi_1|^2 |\phi_{-1}|^2 \right] \mathrm{d}\boldsymbol{x}$$

under constraint $\mathcal{S} := \{(\phi_1, \ \phi_{-1}) \big| \ \|\phi_1\|^2 = \frac{1+M}{2}, \ \|\phi_{-1}\|^2 = \frac{1-M}{2}, \ E_0(\phi_1, \phi_{-1}) < \infty\}$.

THEOREM 2.9 (nullspace of $\boldsymbol{\mathcal{H}_-}$ for antiferromagnetic system). *Under conditions* (2.22)*, then we have the following property for the nullspace of* $\mathcal{H}_-$*:*

$$\mathrm{span}\{\Phi_1, \Phi_2\} \subset \mathrm{null}(\mathcal{H}_-),$$

where $\Phi_1 = \Phi_g$, $\Phi_2 = (b_1 \phi_1^g, 0, b_{-1} \phi_{-1}^g)^\top$, *and* $(b_1, b_{-1}) = \big(-(\frac{1-M}{1+M})^{1/2}, (\frac{1+M}{1-M})^{1/2} \big)$.

Proof. Noticing the fact that $\phi_0 = 0$, we can reduce (1.5) as follows:

$$\begin{cases} [L - \mu_1 + \beta_n \rho + \beta_s (|\phi_1^g|^2 - |\phi_{-1}^g|^2)] \ \phi_1^g = 0, \\ [L - \mu_{-1} + \beta_n \rho - \beta_s (|\phi_1^g|^2 - |\phi_{-1}^g|^2)] \ \phi_{-1}^g = 0, \end{cases} \tag{2.23}$$

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.

Downloaded 07/11/26 to 202.113.11.53 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

and the operator $\mathcal{H}_-$ is simplified below:

$$\mathcal{H}_- = \operatorname{diag}\Big(L-\mu_1+\beta_n\rho+\beta_s\big(|\phi_1^g|^2-|\phi_{-1}^g|^2\big),\quad L-\mu_0+\beta_n\rho+\beta_s\big(\phi_1^g-\phi_{-1}^g\big)^2,\\ L-\mu_{-1}+\beta_n\rho-\beta_s\big(|\phi_1^g|^2-|\phi_{-1}^g|^2\big)\Big).$$

Then, according to (2.23), we obtain that Φ_1 and Φ_2 lie in the nullspace of $\mathcal{H}_-$, i.e., $\mathcal{H}_-\Phi_1 = 0$ and $\mathcal{H}_-\Phi_2 = 0$. It is easy to check that $\langle\Phi_1,\Phi_2\rangle = 0$ and $\|\Phi_2\| = 1$ with $b_1 = -(\frac{1-M}{1+M})^{1/2},\ b_{-1} = (\frac{1+M}{1-M})^{1/2}$. ∎

Remark 2.10. From our extensive numerical results, not fully shown here, we conjecture that the following property holds for antiferromagnetic systems, that is,

$$\mathcal{H}_+ \text{ is invertible} \qquad \text{and} \qquad \operatorname{null}(\mathcal{H}_-) = \operatorname{span}\{\Phi_1,\Phi_2\}.$$

Moreover, $\mathcal{H}_+$ is positive definite and $\mathcal{H}_-$ is positive semidefinite.

If the above property is true, we can obtain the generalized nullspace of $\mathcal{H}$ as follows:

$$\operatorname{null}(\mathcal{H}^3) = \operatorname{null}(\mathcal{H}^2) = \operatorname{null}(\mathcal{H}) \oplus \operatorname{span}\left\{\begin{bmatrix}\widehat{\Phi}_1\\ \mathbf{0}\end{bmatrix},\begin{bmatrix}\widehat{\Phi}_2\\ \mathbf{0}\end{bmatrix}\right\}, \tag{2.24}$$

where $\widehat{\Phi}_j = \mathcal{H}_+^{-1}\Phi_j,\ j = 1,2$, and

$$\operatorname{null}(\mathcal{H}) = \operatorname{span}\left\{\begin{bmatrix}\mathbf{0}\\ \Phi_1\end{bmatrix},\begin{bmatrix}\mathbf{0}\\ \Phi_2\end{bmatrix}\right\}.$$

The convergence of the nonzero eigenvalues depends heavily on the approximation accuracy of the generalized nullspace that is associated with $\mathcal{H}$ [23]. Based on Theorem 2.7 and Remark 2.10, we obtain the generalized nullspace of $\mathcal{H}$, thus essentially improving the convergence and efficiency.

3. Numerical method. In this section, we will propose an efficient and spectrally accurate numerical method to solve the BdG equations (2.14). Due to the presence of external trapping potential $V(\boldsymbol{x})$, the ground states $\Phi_g(\boldsymbol{x})$ and the eigenfunctions $(\boldsymbol{u},\boldsymbol{v})/(\boldsymbol{f},\boldsymbol{g})$ are all smooth and fast decaying. Therefore, it is reasonable to truncate the whole space $\mathbb{R}^d$ into a bounded domain $\Omega\subset\mathbb{R}^d$ that is large enough such that the truncation error is negligible. Since all related functions are smooth, the Fourier spectral (FS) discretization stands out as the optimal candidate for spatial discretization [6, 7, 35], due to its simplicity, spectral accuracy, and great efficiency that is guaranteed by the discrete fast Fourier transform (FFT). The computation domain is usually chosen as a rectangle, denoted as $\mathcal{D}_L := [-L,L]^d$. Spatial discretization details will be presented in the next subsection.

3.1. Spatial discretization by Fourier spectral method. Provided that the stationary states Φ_g and all chemical potentials are precomputed with a very fine mesh by the gradient flow with Lagrange multiplier (GFLM) method [24], the numerical accuracy approaches machine precision. For simplicity, we choose to illustrate the spatial discretization for the 1D case, and extensions to higher dimensions are omitted here. We choose the computational domain $\mathcal{D}_L := [-L,L]$ and discretize it uniformly with mesh size $h_x = \frac{2L}{N}$, where N is a positive even integer. Define the grid point set as $\mathcal{T}_x = \{(-N/2,\dots,N/2)h_x\}$, and introduce the plane wave basis and discrete space as

$$W_k(x) = e^{\mathrm{i}\mu_k(x+L)}\quad(-N/2\le k\le N/2-1)\text{ and }X_N = \operatorname{span}\{W_k(x)\}_{k=-N/2}^{N/2-1},$$

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.

Downloaded 07/11/26 to 202.113.11.53 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

with $\mu_k = \pi k/L$. Define $F(x_n)$ ($F = \phi_j^g, u_j, v_j, V$, etc.) as function value at grid point $x_n = -L + nh_x \in \mathcal{T}_{\boldsymbol{x}}$ and $\mathbf{F} = \big(F(x_1), F(x_2), \ldots, F(x_N)\big)$ as the corresponding discrete vector. The FS approximations of F, denoted by F_N, and its Laplacian $\nabla^2 F$ read as follows:

$$F(x) \approx (\mathcal{I}_N F)(x) := \sum_{k=-N/2}^{N/2-1} \widetilde{F}_k \, W_k(x),$$

$$(\nabla^2 F)(x) \approx (\nabla^2 (\mathcal{I}_N F))(x) = \sum_{k=-N/2}^{N/2-1} -\mu_k^2 \, \widetilde{F}_k \, W_k(x),$$

where $\mathcal{I}_N$ denotes a mapping from $C(\Omega)$ to X_N, and $\widetilde{F}_k$, the discrete Fourier transform of vector $\mathbf{F}$, is computed as

$$\widetilde{F}_k = \frac{1}{N} \sum_{n=0}^{N-1} F(x_n) \overline{W}_k(x_n) = \frac{1}{N} \sum_{n=0}^{N-1} F(x_n) \; e^{\frac{-i2\pi nk}{N}}, \qquad -N/2 \le k \le N/2 - 1 \tag{3.1}$$

and is accelerated by a discrete FFT within $O(N \ln(N))$ float operations. The numerical approximation of the Laplacian operator ∇^2, denoted as $[\![\nabla^2]\!]$, corresponds to a dense matrix [30, 35, 39]. We map the function's pointwise multiplication $F_n G_n$ as a matrix-vector product $[\![F]\!]\mathbf{G}$, that is,

$$\big([\![F]\!]\mathbf{G}\big)_n := F_n G_n \quad \Longrightarrow \quad [\![F]\!] = \operatorname{diag}(\mathbf{F}). \tag{3.2}$$

Therefore, operators $\mathcal{A}$ and $\mathcal{B}$ are mapped into matrices $\mathbf{A}$ and $\mathbf{B}$ in the following way:

(3.3)

$$\mathbf{A} := [\![L\mathbf{I}_3 - \Lambda]\!] + \beta_n \Big([\![\rho \mathbf{I}_3]\!] + [\![\Phi_g \Phi_g^\top]\!] \Big) + \beta_s \Bigg([\![\mathbf{S} \cdot \mathbf{s}(\Phi_g)]\!] + \sum_{j=x,y,z} [\![\mathbf{S}_j \Phi_g \Phi_g^\top \mathbf{S}_j]\!] \Bigg),$$

(3.4)

$$\mathbf{B} := \beta_n [\![\Phi_g \Phi_g^\top]\!] + \beta_s \sum_{j=x,y,z} [\![\mathbf{S}_j \Phi_g \Phi_g^\top \mathbf{S}_j^\top]\!].$$

Thanks to the FS discretization, both matrices $\mathbf{A}$ and $\mathbf{B}$ are symmetric, thus keeping their continuous operators' Hermitian property. Setting $\boldsymbol{u}_N = (\boldsymbol{u}_{1,N}; \boldsymbol{u}_{0,N}; \boldsymbol{u}_{-1,N})$ and $\boldsymbol{v}_N = (\boldsymbol{v}_{1,N}; \boldsymbol{v}_{0,N}; \boldsymbol{v}_{-1,N})$, the BdG equations (2.12) are then discretized into a linear eigenvalue problem:

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ -\mathbf{B} & -\mathbf{A} \end{bmatrix} \begin{bmatrix} \boldsymbol{u}_N \\ \boldsymbol{v}_N \end{bmatrix} = \omega_N \begin{bmatrix} \boldsymbol{u}_N \\ \boldsymbol{v}_N \end{bmatrix}. \tag{3.5}$$

With a similar change of variables, i.e.,

$$\boldsymbol{f}_N = \frac{1}{2}(\boldsymbol{u}_N + \boldsymbol{v}_N), \; \boldsymbol{g}_N = \frac{1}{2}(\boldsymbol{u}_N - \boldsymbol{v}_N), \tag{3.6}$$

the above equation is rewritten into a linear response eigenvalue problem:

$$\begin{bmatrix} \mathbf{O} & \mathbf{H}_- \\ \mathbf{H}_+ & \mathbf{O} \end{bmatrix} \begin{bmatrix} \boldsymbol{f}_N \\ \boldsymbol{g}_N \end{bmatrix} = \omega_N \begin{bmatrix} \boldsymbol{f}_N \\ \boldsymbol{g}_N \end{bmatrix}, \tag{3.7}$$

where $\mathbf{H}_- = \mathbf{A} - \mathbf{B}$ and $\mathbf{H}_+ = \mathbf{A} + \mathbf{B}$.

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.

Downloaded 07/11/26 to 202.113.11.53 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

The above discrete dense eigensystem (3.7) is solved using the recently developed Bi-Orthogonal Structure Preserving algorithm (BOSP) [23], and details are illustrated in the coming subsection.

Remark 3.1 (matching eigenfunction constraint (2.15)). In practice, we *do not* need to bother about the constraint (2.15) when solving the eigenvalue problem (3.7). Actually, once the eigenvectors are computed, the constraint (2.15) will be satisfied easily with a scalar scaling of $\boldsymbol{f}_N$ and $\boldsymbol{g}_N$.

3.2. Bi-orthogonal structure-preserving Fourier spectral eigensolver. The ARPACK package has been successfully adapted to BdG equations of dipolar BECs [22, 35]. However, a similar adaptation attempt is not as effective, and sometimes it does not even converge within a reasonable time, because the generalized nullspace is much larger and more complicated, not to even mention the large-scale dense eigensystem for three-dimensional problems. The eigenspaces associated with eigenvalues of different magnitudes are bi-orthogonal, and such bi-orthogonality property shall be taken into account in the eigensolver design [3, 4, 23]. The structure of generalized nullspace and bi-orthogonality, well described in section 2 on the continuous level, are assessed numerically by the linear algebra package (LAPACK) and a modified Gram–Schmidt bi-orthogonal algorithm. Based on the recently developed linear response eigensolver BOSP [23], by combining the FS method for spatial approximation and making use of the specific generalized nullspace structure (section 2.3), we propose an efficient iterative subspace eigensolver and provide a friendly interface for matrix-vector product evaluation. Since there is no explicit matrix storage and the matrix-vector product is implemented via FFT within almost optimal $O(\mathtt{DOF}\log(\mathtt{DOF}))$ operations ($\mathtt{DOF}$ is the degree of freedom/total number of grid points, i.e., $\mathtt{DOF} = N^d$), the heavy memory burden is much more alleviated, and the computational efficiency is guaranteed to a large extent. Therefore, it provides a feasible solution to large-scale problems, especially in such a densely populated system, as in our case. Thanks to the FS method, our solver can achieve spectral accuracy for both eigenvalue and eigenvectors as long as the BOSP's accuracy tolerance is chosen sufficiently small.

As is known, the matrix-vector product evaluation is the most time-consuming part. In this article, the matrix-vector product

$$\mathbf{H}_+\boldsymbol{f}_N, \quad \mathbf{H}_-\boldsymbol{g}_N \quad \text{for} \quad \boldsymbol{f}_N, \boldsymbol{g}_N \in \mathbb{R}^{3N}$$

can be realized by two pairs of FFT/iFFT plus some function multiplication in physical space. The overall computational costs to compute the first $\mathtt{nev}$ eigenpairs amount to

$$\mathcal{O}(\mathtt{nev}) \times \big(\mathcal{O}(\mathtt{DOF}\log(\mathtt{DOF})) + \mathcal{O}(\mathtt{DOF})\big) + \mathcal{O}(\mathtt{nev}^3) = \mathcal{O}(\mathtt{nev}\ N\log(\mathtt{DOF})) + \mathcal{O}(\mathtt{nev}^3).$$

When the degree of freedom is much larger than the number of eigenvalues, i.e., $\mathtt{DOF} \gg \mathtt{nev}$, the computational costs are approximately $\mathcal{O}(\mathtt{nev} \times \mathtt{DOF}\log(\mathtt{DOF}))$ and will be verified numerically in the next section.

3.3. Stability analysis. To demonstrate the practicality of our solver, we will present a rigorous stability analysis for the eigenfunctions in this subsection. Since the density and eigenfunction both decay fast enough in space, it is reasonable to assume that $\chi(\boldsymbol{x})$ ($\chi = \phi_j^g, u_j, v_j$, etc.) is numerically compactly supported in a bounded domain $\Omega \subsetneq \mathbb{R}^d$, that is, $\mathrm{supp}\{\chi\} \subsetneq \Omega$. We introduce $2L$-periodic Sobolev space $H_p^m(\Omega) \subset H^m(\Omega)$ $(m \geq 1)$ with $\Omega = \mathcal{D}_L$ and the seminorm, norm, and ∞-norms as follows, respectively:

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.

$$|\chi|_m := \left(\sum_{|\boldsymbol{\alpha}|=m} \|\partial^{\boldsymbol{\alpha}}\chi\|^2\right)^{1/2}, \quad \|\chi\|_m := \left(\sum_{|\boldsymbol{\alpha}|\le m} \|\partial^{\boldsymbol{\alpha}}\chi\|^2\right)^{1/2}, \quad \|\chi\|_\infty = \sup_{\boldsymbol{x}\in\Omega} |\chi(\boldsymbol{x})|,$$

with index $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_d) \in \mathbb{Z}^d$, $|\boldsymbol{\alpha}| = \sum_{j=1}^d \alpha_j$, $\partial^{\boldsymbol{\alpha}} = \partial_{x_1}^{\alpha_1} \cdots \partial_{x_d}^{\alpha_d}$, and $\|\cdot\|$ being the L^2-norm. For vector-valued function $\boldsymbol{\chi} = (\chi_1, \chi_0, \chi_{-1})^\top$, we define $|\boldsymbol{\chi}|_m := \sqrt{\sum_{j=-1}^1 |\chi_j|_m^2}$, $\|\boldsymbol{\chi}\|_m := \sqrt{\sum_{j=-1}^1 \|\chi_j\|_m^2}$, $\|\boldsymbol{\chi}\| := \sqrt{\sum_{j=-1}^1 \|\chi_j\|^2}$, and $\|\boldsymbol{\chi}\|_\infty := \max\{\|\chi_j\|_\infty\}$. We use $A \lesssim B$ to denote $A \le cB$, where the constant $c > 0$ is independent of the grid number N. Then we introduce the stability analysis as follows.

THEOREM 3.2 (stability analysis). *For any* $\boldsymbol{\chi}(\boldsymbol{x}) = (\chi_1(\boldsymbol{x}), \chi_0(\boldsymbol{x}), \chi_{-1}(\boldsymbol{x}))^\top \in (H_p^m(\Omega))^3$ *with* $m > d/2 + 2$ *and its FS approximations*

$$(\mathcal{I}_N\boldsymbol{\chi})(\boldsymbol{x}) = \big((\mathcal{I}_N\chi_1)(\boldsymbol{x}), (\mathcal{I}_N\chi_0)(\boldsymbol{x}), (\mathcal{I}_N\chi_{-1})(\boldsymbol{x})\big)^\top,$$

we have the following error estimate:

$$\|\mathcal{Q}\boldsymbol{\chi} - \mathcal{Q}(\mathcal{I}_N\boldsymbol{\chi})\|_\infty \lesssim N^{-(m-\frac{d}{2}-2)}, \tag{3.8}$$

where $\mathcal{Q} = \mathcal{A}, \mathcal{B}$, *and* $\boldsymbol{\chi} = \boldsymbol{u}, \boldsymbol{v}$.

To prove the above theorem, the following preparations are required.

LEMMA 3.3 (Fourier spectral approximation [25, 30]). *For* $\chi(\boldsymbol{x}) \in H_p^m(\Omega)$ *with* $m > d/2$ *and its Fourier spectral approximation* $\mathcal{I}_N\chi$, *we have the following error estimates:*

$$\|\partial^{\boldsymbol{\alpha}}(\chi - \mathcal{I}_N\chi)\|_\infty \lesssim N^{-(m-\frac{d}{2}-|\boldsymbol{\alpha}|)}|\chi|_m, \qquad 0 \le |\boldsymbol{\alpha}| \le m, \tag{3.9}$$

$$\|\partial^{\boldsymbol{\alpha}}(\chi - \mathcal{I}_N\chi)\| \lesssim N^{-(m-|\boldsymbol{\alpha}|)}|\chi|_m, \qquad 0 \le |\boldsymbol{\alpha}| \le m. \tag{3.10}$$

Since operators $\mathcal{A}$ and $\mathcal{B}$ contain only two types of operators, namely the Laplacian operator ∇^2 and the function multiplication operation that involves $V(\boldsymbol{x})$ & $\phi_i^g\phi_j^g$, the following corollary is a prerequisite for proving Theorem 3.2.

COROLLARY 3.4. *For* $\chi(\boldsymbol{x}) \in H_p^m(\Omega)$ *with* $m > d/2$ *and its Fourier spectral approximation* $\mathcal{I}_N\chi$, *we have*

$$\|\nabla^2\chi - \nabla^2(\mathcal{I}_N\chi)\|_\infty \lesssim N^{-(m-\frac{d}{2}-2)}|\chi|_m, \tag{3.11}$$

$$\|V\chi - V(\mathcal{I}_N\chi)\|_\infty \lesssim N^{-(m-\frac{d}{2})}|\chi|_m, \tag{3.12}$$

$$\|\phi_i^g\phi_j^g\chi - \phi_i^g\phi_j^g(\mathcal{I}_N\chi)\|_\infty \lesssim N^{-(m-\frac{d}{2})}|\chi|_m \quad (i,j = 1, 0, -1). \tag{3.13}$$

Since it is easy to prove the above corollary using Lemma 3.3, we shall omit details for brevity.

The proof of Theorem 3.2

Proof. For simplicity, we only prove the $\mathcal{Q} = \mathcal{A}$ case, and extensions to other operators are similar. From (3.9)–(3.12), we obtain

$$\|(L\mathbf{I}_3 - \Lambda)\boldsymbol{\chi} - (L\mathbf{I}_3 - \Lambda)(\mathcal{I}_N\boldsymbol{\chi})\|_\infty \lesssim N^{-(m-\frac{d}{2}-2)}|\boldsymbol{\chi}|_m.$$

For the second and third terms of $\mathcal{A}$ as shown in (2.5) (the same as $\mathbf{A}$ in (3.3)), using (3.9), (3.13), and the triangle inequality, the following error estimate holds:

$$\big\|\big(\mathcal{A} - (L\mathbf{I}_3 - \Lambda)\big)\boldsymbol{\chi} - \big(\mathcal{A} - (L\mathbf{I}_3 - \Lambda)\big)(\mathcal{I}_N\boldsymbol{\chi})\big\|_\infty \lesssim N^{-(m-\frac{d}{2})}|\boldsymbol{\chi}|_m.$$

Therefore, we finish the proof of Theorem 3.2. ∎

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.

Downloaded 07/11/26 to 202.113.11.53 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

Downloaded 07/11/26 to 202.113.11.53 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

3.4. Convergence analysis. In this subsection, we will propose the convergence analysis of FS approximations. To begin with, we adopt the Sobolev space $\mathbb{V} = (H_p^1(\Omega))^3 \times (H_p^1(\Omega))^3$ with the following norm:

$$\|\mathbf{\Phi}\|_1 = \sqrt{\|\boldsymbol{f}\|_1^2 + \|\boldsymbol{g}\|_1^2} \quad \text{and} \quad \|\mathbf{\Phi}\| = \sqrt{\|\boldsymbol{f}\|^2 + \|\boldsymbol{g}\|^2} \qquad \forall\, \mathbf{\Phi} := (\boldsymbol{f}; \boldsymbol{g}) \in \mathbb{V}.$$

The BdG equation (2.14) is equivalent to the following eigenvalue problem:

$$\widetilde{\mathcal{H}} \begin{bmatrix} \boldsymbol{f} \\ \boldsymbol{g} \end{bmatrix} := \begin{bmatrix} \mathcal{H}_+ & \mathcal{O} \\ \mathcal{O} & \mathcal{H}_- \end{bmatrix} \begin{bmatrix} \boldsymbol{f} \\ \boldsymbol{g} \end{bmatrix} = \omega \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \boldsymbol{f} \\ \boldsymbol{g} \end{bmatrix}. \tag{3.14}$$

The Galerkin weak form of (3.14) reads as follows: find $0 \neq \omega \in \mathbb{R}$ and $\mathbf{\Phi} = (\boldsymbol{f}; \boldsymbol{g}) \in \mathbb{V}$ such that

$$a(\mathbf{\Phi}, \mathbf{\Psi}) = \omega\, b(\mathbf{\Phi}, \mathbf{\Psi}) \quad \forall\, \mathbf{\Psi} = (\boldsymbol{\xi}; \boldsymbol{\eta}) \in \mathbb{V}, \tag{3.15}$$

subject to constraint $b(\mathbf{\Phi}, \mathbf{\Phi}) = 1/2$ with

$$a(\mathbf{\Phi}, \mathbf{\Psi}) := \langle \mathcal{H}_+ \boldsymbol{f}, \boldsymbol{\xi} \rangle + \langle \mathcal{H}_- \boldsymbol{g}, \boldsymbol{\eta} \rangle, \quad b(\mathbf{\Phi}, \mathbf{\Psi}) := \langle \boldsymbol{g}, \boldsymbol{\xi} \rangle + \langle \boldsymbol{f}, \boldsymbol{\eta} \rangle,$$

where $\langle \boldsymbol{f}, \boldsymbol{g} \rangle := \sum_{j=-1}^{1} \int_{\mathbb{R}^d} f_j(\boldsymbol{x}) \overline{g_j(\boldsymbol{x})}\, \mathrm{d}\boldsymbol{x}$ $\forall \boldsymbol{f}, \boldsymbol{g} \in (L^2(\Omega))^3$.

Remark 3.5. From Theorem 2.7 or Remark 2.10, we can prove that there exists $c_0 > 0$ such that

$$a(\mathbf{\Phi}, \mathbf{\Phi}) \geq c_0 \|\mathbf{\Phi}\|_1 \quad \forall\, \mathbf{\Phi} \in \mathbb{U}, \tag{3.16}$$

where the subspace $\mathbb{U}$ is defined as

$$\mathbb{U} = \mathrm{null}(\mathcal{H}_+)^\perp \times \mathrm{null}(\mathcal{H}_-)^\perp \subset \mathbb{V},$$

with $\mathrm{null}(\mathcal{H}_\pm)^\perp := \{\boldsymbol{f} \in (H_p^1(\Omega))^3 \,|\, \langle \boldsymbol{f}, \boldsymbol{\xi} \rangle = 0\ \forall\, \boldsymbol{\xi} \in \mathrm{null}(\mathcal{H}_\pm)\}$. It is important to point out that solving the BdG equation (3.15) in $\mathbb{U}$ is equivalent to solving nonzero ω and its associated eigenfunction $\mathbf{\Phi}$ in $\mathbb{V}$.

Define the approximation finite-dimensional spaces X_N of $H_p^1(\Omega)$ and $\mathbb{V}_N$ of $\mathbb{V}$ as follows:

$$X_N := \mathrm{span}\,\{W_k(\boldsymbol{x}), k = -N/2, \ldots, N/2 - 1\}, \quad \mathbb{V}_N := (X_N)^3 \times (X_N)^3,$$

and consider its approximation problem: find $0 \neq \omega_N \in \mathbb{R}$ and $\mathbf{\Phi}_N = (\boldsymbol{f}_N; \boldsymbol{g}_N) \in \mathbb{V}_N$ such that

$$a_N(\mathbf{\Phi}_N, \mathbf{\Psi}_N) = \omega_N\, b_N(\mathbf{\Phi}_N, \mathbf{\Psi}_N) \quad \forall\, \mathbf{\Psi}_N = (\boldsymbol{\xi}_N; \boldsymbol{\eta}_N) \in \mathbb{V}_N, \tag{3.17}$$

subject to normalization constraint $b_N(\mathbf{\Phi}_N, \mathbf{\Phi}_N) = 1/2$ with

$$\begin{aligned} a_N(\mathbf{\Phi}_N, \mathbf{\Psi}_N) &= \langle \mathcal{H}_+ \boldsymbol{f}_N, \boldsymbol{\xi}_N \rangle_N + \langle \mathcal{H}_- \boldsymbol{g}_N, \boldsymbol{\eta}_N \rangle_N, \\ b_N(\mathbf{\Phi}_N, \mathbf{\Psi}_N) &= \langle \boldsymbol{g}_N, \boldsymbol{\xi}_N \rangle_N + \langle \boldsymbol{f}_N, \boldsymbol{\eta}_N \rangle_N, \end{aligned}$$

and $\langle \boldsymbol{f}, \boldsymbol{\xi} \rangle_N := \sum_{j=-1}^{1} \langle f_j^N, \xi_j^N \rangle_N := \sum_{j=-1}^{1} \big(\frac{2L}{N} \sum_{n=0}^{N} f_j^N(\boldsymbol{x}_n) \overline{g_j^N(\boldsymbol{x}_n)} \big)$.
Similar to Lemma A.1, we have the following equivalence result and choose to omit proofs for brevity.

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.

Downloaded 07/11/26 to 202.113.11.53 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

LEMMA 3.6. *The discrete problem* (3.7) *and the discrete variational problem* (3.17) *are equivalent.*

To illustrate the convergence behavior, we introduce the following notation:

$$\delta_N(\mathbf{\Phi}) = \inf_{\mathbf{\Psi}\in\mathbb{V}_N} \left\{ \|\mathbf{\Phi}-\mathbf{\Psi}\|_1 + \sup_{\mathbf{\Theta}\in\mathbb{V}_N} \frac{|a(\mathbf{\Psi},\mathbf{\Theta}) - a_N(\mathbf{\Psi},\mathbf{\Theta})|}{\|\mathbf{\Theta}\|_1} \right\}. \tag{3.18}$$

Similar to Lemma A.2 for a scalar function, by the coerciveness of bilinear operator $a(\cdot,\cdot)$ (see (3.16)) and the conformal subspace approximation, we obtain

$$\delta_N(\mathbf{\Phi}) \lesssim N^{-(m-\sigma)}, \qquad \text{with} \quad \sigma = \max\{1, d/2\}. \tag{3.19}$$

According to the general theory given by Lemma A.3, we derive the following error estimates.

LEMMA 3.7. *For any eigenpair approximation* $\{\omega_N; \mathbf{\Phi}_N\}$ *of* (3.17)*, there is an eigenpair* $\{\omega; \mathbf{\Phi}\}$ *of* (3.15) *corresponding to* ω *such that*

$$\|\mathbf{\Phi}-\mathbf{\Phi}_N\|_1 \lesssim \delta_N(\mathbf{\Phi}), \quad \|\mathbf{\Phi}-\mathbf{\Phi}_N\| \lesssim \zeta_N \|\mathbf{\Phi}-\mathbf{\Phi}_N\|_1, \quad |\omega-\omega_N| \lesssim \|\mathbf{\Phi}-\mathbf{\Phi}_N\|_1^2,$$

where the constant ζ_N *approaches* 0 *as* $N\to\infty$.

Finally, via a change of variables, i.e., (2.13) and (3.6), we obtain error estimates for $(\boldsymbol{u}, \boldsymbol{v})$.

THEOREM 3.8 (error estimates). *If* $\boldsymbol{u}, \boldsymbol{v} \in (H_p^m(\Omega))^3$ *with* $m \geq 1$ *and* $\mathrm{supp}\{\boldsymbol{u}\}$, $\mathrm{supp}\{\boldsymbol{v}\} \subsetneq \Omega$*, then for any eigenpair approximation* $\{\omega_N; \boldsymbol{u}_N, \boldsymbol{v}_N\}$ *of* (3.5)*, there is an eigenpair* $\{\omega; \boldsymbol{u}, \boldsymbol{v}\}$ *of* (2.12) *satisfying the following error estimates:*

$$\|\boldsymbol{u}-\boldsymbol{u}_N\| + \|\boldsymbol{v}-\boldsymbol{v}_N\| \lesssim N^{-(m-\sigma)},$$
$$|\omega-\omega_N| \lesssim N^{-2(m-\sigma)},$$

where $\sigma = \max\{1, d/2\}$.

4. Numerical results. In this section, we first carry out comprehensive numerical studies to illustrate the accuracy and efficiency of our solver. Then we apply it to investigate the Bogoliubov excitations around the ground states of spin-1 BECs. The ground states Φ_g and the chemical potentials are precomputed with accuracy close to machine precision via the GFLM method [24] in a large enough domain with a small enough mesh size. Unless stated otherwise, we choose the potential $V(\boldsymbol{x})$ as the harmonic trapping potential (1.2) and set the computational domain as squares and $L = 16$ for 1D and 2D and $L = 8$ for 3D, respectively, i.e., $\mathcal{D} = [-16, 16]$ in 1D, $\mathcal{D} = [-16, 16]^2$ in 2D, and $\mathcal{D} = [-8, 8]^3$ in 3D, respectively. The domain is discretized uniformly in each spatial direction with the same mesh size $h = 2L/N$ for simplicity. We choose to study the *ferromagnetic* (FM) case with $\beta_n = 885.4$, $\beta_s = -4.1$ and the *antiferromagnetic* (anti-FM) case with $\beta_n = 240.8$, $\beta_s = 7.5$ throughout the whole section.

4.1. Performance investigation. First, we verify the spectral accuracy for different space dimensions. Analytical eigenvalues and eigenvectors were given for the harmonic trapping potential by (2.16) in Theorem 2.3. For an eigenvalue ω with multiplicity k, we denote its associated analytical eigenspaces as $\mathcal{M}_{\boldsymbol{u}} := \mathrm{span}\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_k\}$, $\mathcal{M}_{\boldsymbol{v}} := \mathrm{span}\{\boldsymbol{v}_1, \cdots, \boldsymbol{v}_k\}$ and denote $\{\omega_{\alpha,N}; \boldsymbol{u}_{\alpha,N}, \boldsymbol{v}_{\alpha,N}\}$ as the numerical approximations of eigenvalue ω_α and eigenfunctions $(\boldsymbol{u}_\alpha, \boldsymbol{v}_\alpha)$ obtained with mesh size $h = 2L/N$. To demonstrate the convergence, we adopt the following error functions:

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.

$$e^h_{\omega_\alpha} := \frac{|\omega_{\alpha,N} - \omega_\alpha|}{|\omega_\alpha|}, \qquad e^{h,\alpha}_{\boldsymbol{uv}} := \frac{\|\boldsymbol{u}_{\alpha,N} - \mathcal{P}_{\boldsymbol{u}}\boldsymbol{u}_{\alpha,N}\|_2}{\|\boldsymbol{u}_{\alpha,N}\|_2} + \frac{\|\boldsymbol{v}_{\alpha,N} - \mathcal{P}_{\boldsymbol{v}}\boldsymbol{v}_{\alpha,N}\|_2}{\|\boldsymbol{v}_{\alpha,N}\|_2},$$

where $\alpha = x$ in 1D, $\alpha = x, y$ in 2D, and $\alpha = x, y, z$ in 3D, $\|\cdot\|_2$ is the discrete l^2-norm, and $\mathcal{P}_\nu$ ($\nu = \boldsymbol{u}, \boldsymbol{v}$) is the l^2-orthogonal projection operator into space $\mathcal{M}_\nu$. In both examples, we compute the first 40 eigenvalues and their eigenfunctions.

Example 4.1 (accuracy). We study the accuracy convergence for both the FM and the anti-FM cases in 1D/2D/3D. To this end, we consider the following four cases:

Case I. 1D case: $\gamma_x = 1$.
Case II. Isotropic 2D case: $\gamma_x = \gamma_y = 1$.
Case III. Anisotropic 2D case: $\gamma_x = \gamma_y/2 = 1$.
Case IV. Isotropic 3D case: $\gamma_x = \gamma_y = \gamma_z = 1$.

For **Case I**, there exists one analytical eigenvalue $\omega_x = 1$ with multiplicity $k = 1$. For **Case II**, there exist two analytical eigenvalues $\omega_x = \omega_y = 1$ with multiplicity $k = 2$. For **Case III**, $\omega_x = 1$ and $\omega_y = 2$ are eigenvalues with the same multiplicity $k = 1$. Similarly, for **Case IV**, there exist three analytical eigenvalues $\omega_x = \omega_y = \omega_z = 1$ with $k = 3$. Tables 1–4 illustrate the numerical errors of eigenvalues and eigenvectors computed with different mesh sizes h in **Cases I–IV**.

For the anti-FM case in **Case I**, the convergence rates are compared between our proposed algorithm and LOBP4dCG [3, 4]. To address the positive semidefiniteness of the operator $\mathcal{H}_-$, a shift technique is employed for LOBP4dCG with a shift value of 10 to achieve optimal convergence performance. Figure 1 shows the residual decay curves of the first six smallest eigenvalues $(\lambda_1, \ldots, \lambda_6)$ corresponding to the two algorithms. It can be seen that our algorithm converges faster than LOBP4dCG.

Example 4.2 (efficiency). We investigate the efficiency performance using different mesh sizes for FM and anti-FM cases in 2D and 3D. To this end, we consider the following cases:

TABLE 1
Errors of the eigenvalue/eigenvector for ***Case*** **I** *in* **1*D*** *for* ***FM*** *(upper) and* ***anti-FM*** *cases (lower) in Example* 4.1.

		$h_0 = 1$	$h_0/2$	$h_0/4$	$h_0/8$	$h_0/16$
FM	$e^h_{\omega_x}$	4.392E-03	9.570E-05	2.422E-08	1.383E-09	1.648E-12
	$e^{h,x}_{\boldsymbol{uv}}$	1.069E-01	1.234E-03	2.911E-06	4.481E-08	1.165E-12
anti-FM	$e^h_{\omega_x}$	6.103E-02	1.831E-04	3.283E-10	6.677E-13	3.020E-14
	$e^{h,x}_{\boldsymbol{uv}}$	9.106E-01	1.156E-03	1.027E-06	1.609E-12	2.677E-12

TABLE 2
Errors of the eigenvalue/eigenvector for ***Case*** **II** *in* **2*D*** *for* ***FM*** *(upper) and* ***anti-FM*** *cases (lower) in Example* 4.1.

		$h_0 = 1$	$h_0/2$	$h_0/4$	$h_0/8$	$h_0/16$
FM	$e^h_{\omega_x}$	6.022E-03	1.685E-05	7.811E-11	7.896E-13	5.014E-13
	$e^h_{\omega_y}$	6.022E-03	1.685E-05	7.812E-11	8.007E-13	4.925E-13
	$e^{h,x}_{\boldsymbol{uv}}$	3.503E-02	8.927E-04	4.250E-07	4.627E-08	2.751E-10
	$e^{h,y}_{\boldsymbol{uv}}$	3.506E-02	8.928E-04	4.250E-07	2.625E-08	2.751E-10
anti-FM	$e^h_{\omega_x}$	6.825E-03	1.907E-05	4.664E-10	1.281E-12	4.241E-13
	$e^h_{\omega_y}$	6.825E-03	1.907E-05	2.822E-11	1.789E-11	4.607E-13
	$e^{h,x}_{\boldsymbol{uv}}$	5.013E-02	7.403E-04	1.558E-07	4.135E-10	1.147E-12
	$e^{h,y}_{\boldsymbol{uv}}$	5.006E-02	7.403E-04	1.555E-07	5.368E-10	1.416E-12

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.

Downloaded 07/11/26 to 202.113.11.53 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

Downloaded 07/11/26 to 202.113.11.53 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

TABLE 3

Errors of the eigenvalue/eigenvector for ***Case*** **III** *in* **2*D*** *for* ***FM*** *(upper) and* ***anti-FM*** *cases (lower) in Example* 4.1.

		$h_0 = 1$	$h_0/2$	$h_0/4$	$h_0/8$	$h_0/16$
FM	$e^h_{\omega_x}$	9.823E-03	1.153E-05	2.936E-11	2.138E-10	3.044E-12
	$e^h_{\omega_y}$	2.185E-02	1.514E-03	7.471E-08	6.092E-10	1.598E-12
	$e^{h,x}_{\boldsymbol{uv}}$	2.644E-02	8.649E-04	6.728E-07	1.398E-08	6.398E-09
	$e^{h,y}_{\boldsymbol{uv}}$	1.077	9.706E-03	3.186E-05	4.781E-08	3.306E-09
anti-FM	$e^h_{\omega_x}$	7.222E-03	3.431E-05	1.844E-10	3.888E-13	8.724E-13
	$e^h_{\omega_y}$	4.328E-02	5.590E-04	4.039E-08	1.648E-13	4.867E-13
	$e^{h,x}_{\boldsymbol{uv}}$	3.484E-02	8.149E-04	2.777E-07	1.562E-12	7.409E-11
	$e^{h,y}_{\boldsymbol{uv}}$	1.738E-01	9.128E-03	1.706E-05	4.258E-11	7.208E-12

TABLE 4

Errors of the eigenvalue/eigenvector for ***Case*** **IV** *in* **3*D*** *for* ***FM*** *(upper) and* ***anti-FM*** *cases (lower) in Example* 4.1.

		$h_0 = 1$	$h_0/2$	$h_0/4$	$h_0/8$
FM	$e^h_{\omega_x}$	3.526E-03	6.684E-06	6.108E-11	1.027E-11
	$e^h_{\omega_y}$	3.526E-03	6.684E-06	7.500E-12	1.028E-11
	$e^h_{\omega_z}$	3.526E-03	6.684E-06	4.598E-11	1.028E-11
	$e^{h,x}_{\boldsymbol{uv}}$	4.434E-02	6.697E-04	1.116E-07	4.046E-09
	$e^{h,y}_{\boldsymbol{uv}}$	4.433E-02	6.697E-04	1.116E-07	4.046E-09
	$e^{h,z}_{\boldsymbol{uv}}$	4.443E-02	6.697E-04	1.116E-07	4.046E-09
anti-FM	$e^h_{\omega_x}$	2.061E-03	3.620E-06	1.602E-11	8.528E-13
	$e^h_{\omega_y}$	2.061E-03	3.620E-06	1.213E-12	5.934E-13
	$e^h_{\omega_z}$	2.061E-03	3.620E-06	3.218E-11	5.874E-13
	$e^{h,x}_{\boldsymbol{uv}}$	5.726E-02	5.812E-04	5.043E-08	5.873E-11
	$e^{h,y}_{\boldsymbol{uv}}$	5.712E-02	5.812E-04	5.042E-08	5.873E-11
	$e^{h,z}_{\boldsymbol{uv}}$	5.713E-02	5.812E-04	5.043E-08	5.873E-11

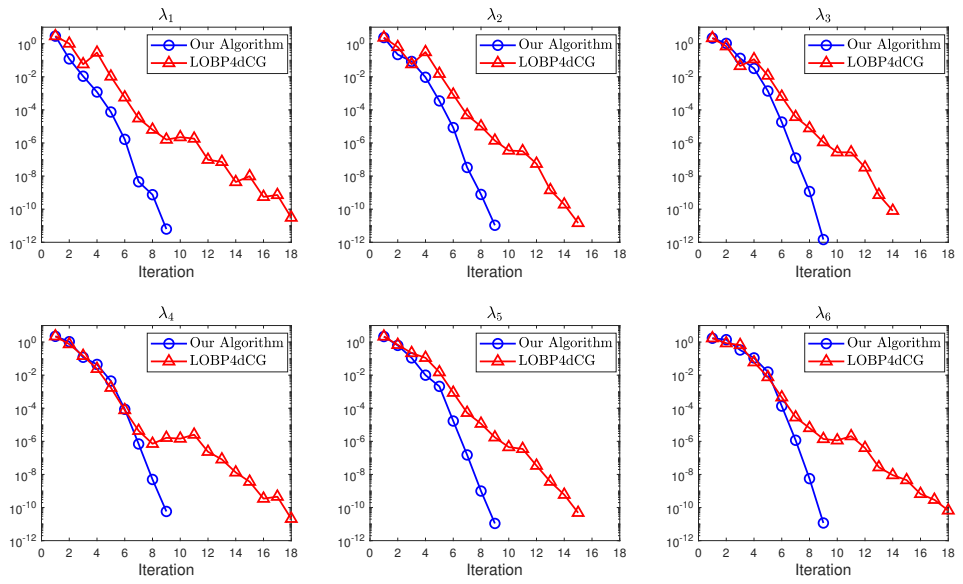


FIG. 1. *Descending trend of the normalized residuals in Example* 4.1.

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.

Case I. Isotropic 2D case: $\gamma_x = \gamma_y = 1$.
Case II. Isotropic 3D case: $\gamma_x = \gamma_y = \gamma_z = 1$.

Figure 2 presents the computation times (measured in seconds) versus the degrees of freedom for 2D and 3D cases. The degree of freedom varies: DOF = 6 144, 24 576, 98 304, 393 216, 1 572 864, 6 291 456 in 2D and DOF = 24 576, 196 608, 1 572 864, 12 582 912 in 3D, while the number of eigenvalues is kept unchanged as nev = 40. It is obvious that the degree of freedom is much larger than the number of eigenvalues, i.e.,

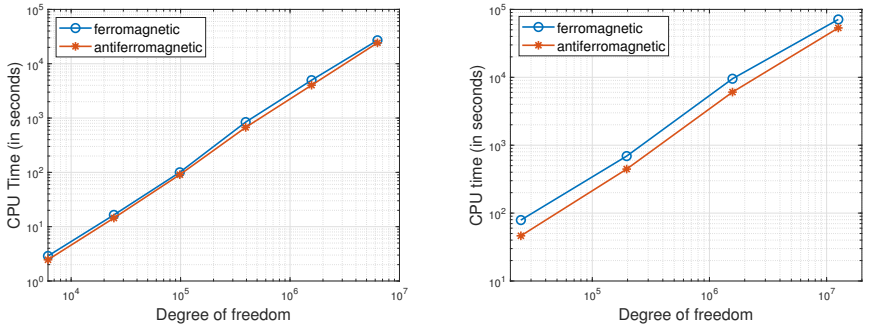


FIG. 2. *The computational times for* ***Case*** **I** *in* **2*D*** *(left) and* ***Case*** **II** *in* **3*D*** *(right) in Example* 4.2.

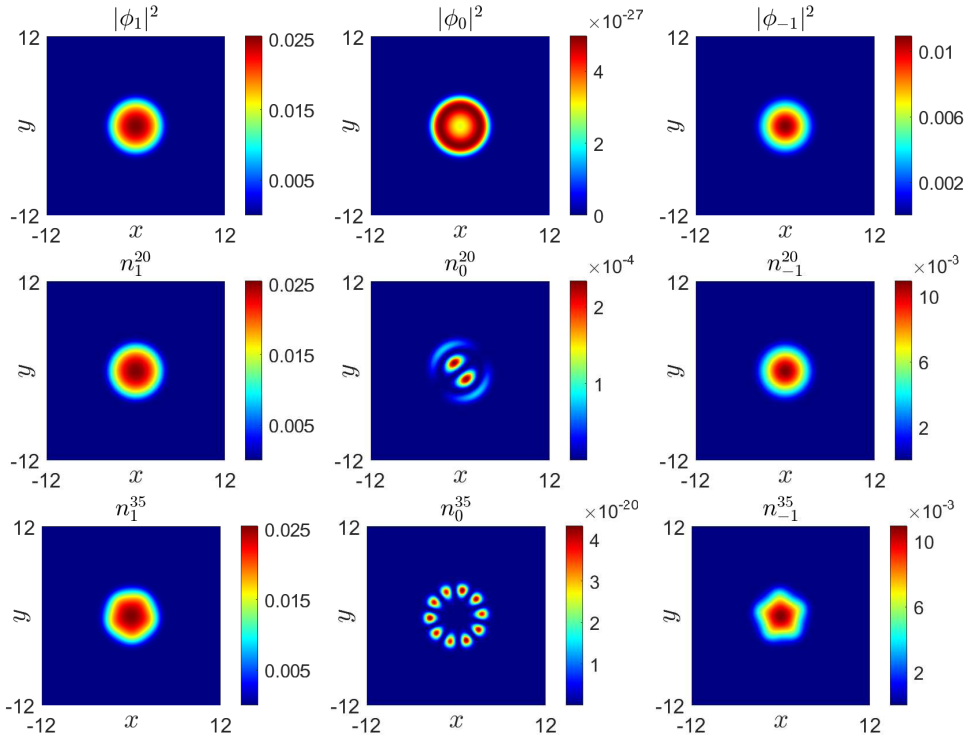


FIG. 3. *Snapshots of* $|\phi_j(\boldsymbol{x})|^2$ *(top row) and perturbed densities* $n_j^\ell(\boldsymbol{x}, t = 10.6)$ *by different excitation modes:* $\ell = 20, 35$ *(middle and bottom rows) for* ***Case*** **I** *in Example* 4.3 *(*$j = 1, 0, -1$ *from left to right).*

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.

Downloaded 07/11/26 to 202.113.11.53 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

Downloaded 07/11/26 to 202.113.11.53 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

$\texttt{DOF} \gg \texttt{nev}$; therefore, the overall computational complexity is around $\mathcal{O}(\texttt{DOF}\log(\texttt{DOF}))$, which can be observed quite clearly from Figure 2. Given the fact that the discrete BdG system is nonsymmetric and dense, and the first $\texttt{nev} = 40$ eigenpairs of the 3D problem can be computed with a **12 million** degree of freedom **within hours** using solely sequentially one-CPU computing, it is reasonable to expect a promising future in physical applications.

4.2. Applications. In this section, we apply our solver to investigate the excitation spectrum and Bogoliubov amplitudes of spin-1 BECs around the ground state. To visualize a normal mode, similar to (2.1), we analyze the evolution of the perturbed density profile [19]:

$$n_j^\ell(\boldsymbol{x},t) = \left|\left[\phi_j(\boldsymbol{x}) + \varepsilon\left(u_j^\ell(\boldsymbol{x})e^{-i\omega_\ell t} + \bar{v}_j^\ell(\boldsymbol{x})e^{i\omega_\ell t}\right)\right]\right|^2, \quad j = 0, \pm 1, \tag{4.1}$$

which reveals the nature of the excitations, with ℓ being the eigenpair index. To this end, we choose a very fine mesh size $h = 1/8$ in both two and three spatial dimensions.

Example 4.3 (2D case). We investigate the perturbed density by different excitation modes in FM and anti-FM condensates in 2D. To this end, we choose $\varepsilon = 0.1$ and consider the following cases:

Case I. Isotropic anti-FM condensates: $\gamma_x = \gamma_y = 1$, $\ell = 20, 35$.

Case II. Anisotropic FM condensates: $\gamma_x = \gamma_y/2 = 1$, $\ell = 30, 55$.

Figures 3–4 display the numerical excitations $n_j^\ell(\boldsymbol{x},t)$ of the BdG equations (at $t = 10.6$) that are associated with the eigenvalues ω_ℓ for **Cases I–II**. From these

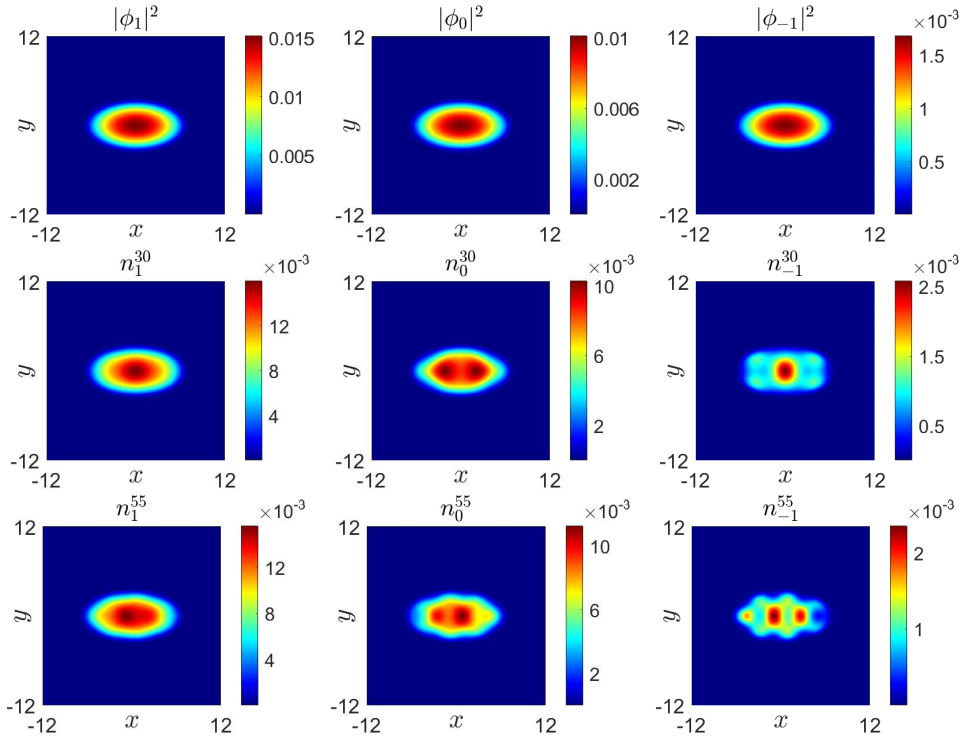


FIG. 4. *Snapshots of* $|\phi_j(\boldsymbol{x})|^2$ *(top row) and perturbed densities* $n_j^\ell(\boldsymbol{x}, t = 10.6)$ *by different excitation modes:* $\ell = 30, 55$ *(middle and bottom rows) for* ***Case*** **II** *in Example* 4.3 *(*$j = 1, 0, -1$ *from left to right).*

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.

figures, we can see that both the spinor and the external potential affect the shape of the excitations essentially and significantly. The eigenmodes $(\boldsymbol{u}^\ell, \boldsymbol{v}^\ell)$ are symmetric or antisymmetric in a symmetric or antisymmetric external potential, respectively. Meanwhile, the eigenmodes will be compressed along the direction with a larger trapping frequency. Indeed, the presence of spinor and anisotropic external potential brings many more rich phase diagrams for eigenmodes of the BdG equations, which will be detailed in the future.

Example 4.4 (3D case). We study the Bogoliubov amplitudes of the BdG equations in FM and anti-FM condensates in 3D. To this end, we study the following four cases:

Case I. Isotropic FM condensates: $\gamma_x = \gamma_y = \gamma_z = 1$, $\ell = 25$.

Case II. Isotropic anti-FM condensates: $\gamma_x = \gamma_y = \gamma_z = 1$, $\ell = 26$.

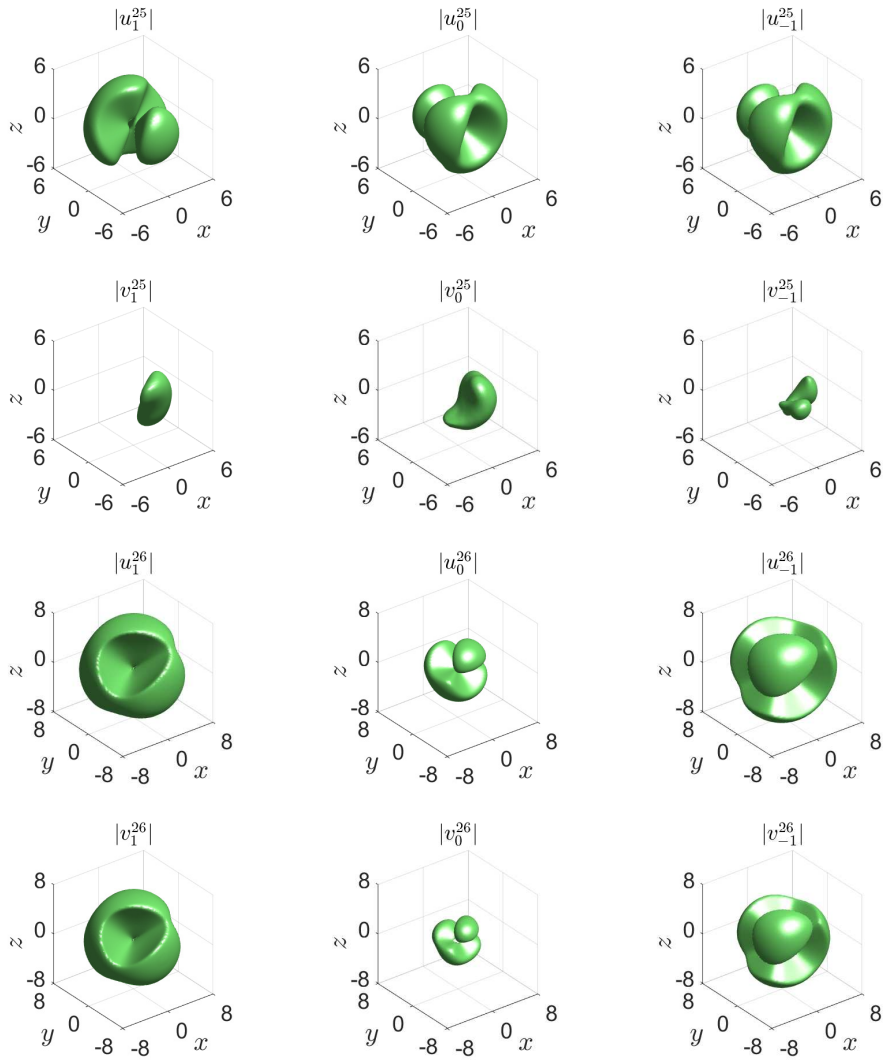


FIG. 5. *In Example* 4.4, *isosurface plots of the Bogoliubov amplitudes for* ***Case*** **I:** $|u_j^{25}| = 10^{-3}$ *(1st row),* $|v_j^{25}| = 10^{-11}$ *(2nd row), and* ***Case*** **II:** $|u_j^{26}| = 10^{-10}$ *(3rd row),* $|v_j^{26}| = 10^{-11}$ *(4th row) (from left to right:* $j = 1, 0, -1$*).*

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.

Downloaded 07/11/26 to 202.113.11.53 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

Case III. Anisotropic FM condensates: $\gamma_x = \gamma_y = \gamma_z/2 = 1$, $\ell = 62$.
Case IV. Anisotropic anti-FM condensates: $\gamma_x = \gamma_y = \gamma_z/2 = 1$, $\ell = 42$.

Figures 5–6 display isosurface plots of the eigenmodes $\boldsymbol{u}^\ell = (u_1^\ell, u_0^\ell, u_{-1}^\ell)^\top$ and $\boldsymbol{v}^\ell = (v_1^\ell, v_0^\ell, v_{-1}^\ell)^\top$ associated with the different ℓ for **Cases I–IV**. From these figures, we can see that the external potential affects the shape of the eigenmodes $\boldsymbol{u}^\ell$ and $\boldsymbol{v}^\ell$ essentially and significantly. The eigenmodes can be symmetric or antisymmetric even in a symmetric trapping potential. Meanwhile, all eigenmodes will be compressed along the direction with a larger trapping frequency. For the 3D case, the presence of spinor and anisotropic external potential brings in many rich phase diagrams for eigenmodes, and we shall leave them as a future study.

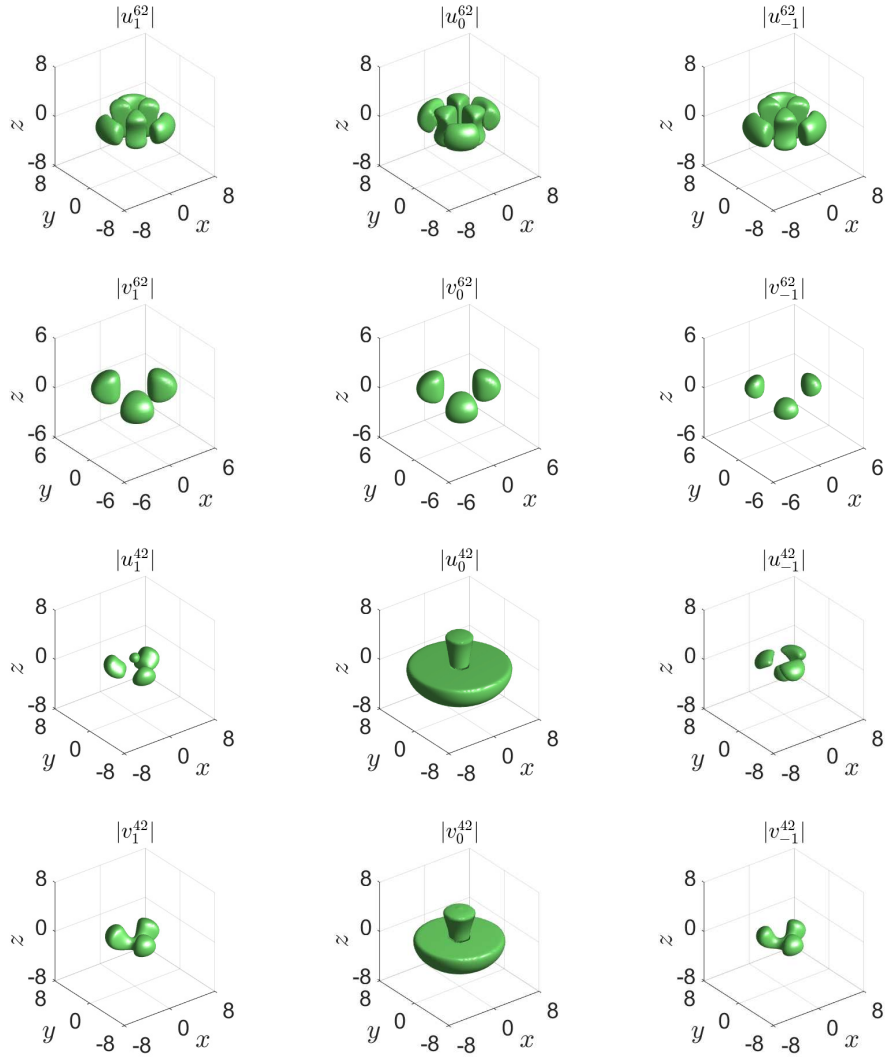


FIG. 6. *In Example* 4.4, *isosurface of the Bogoliubov amplitudes for* ***Case*** **III:** $|u_j^{62}| = 10^{-3}$ *(1st row),* $|v_j^{62}| = 10^{-12}$ *(2nd row), and* ***Case*** **IV:** $|u_j^{42}| = 10^{-9}$ *(3rd row),* $|v_j^{42}| = 10^{-10}$ *(4th row) (from left to right:* $j = 1, 0, -1$*).*

Downloaded 07/11/26 to 202.113.11.53 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.

Downloaded 07/11/26 to 202.113.11.53 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

5. Conclusion. We proposed an efficient and spectrally accurate solver for computing the BdG equations of the spin-1 BEC. We first investigated its analytical eigenpairs, the structure of the generalized nullspace, and the bi-orthogonal property of eigenspaces on a continuous level. Then, based on a stable Gram–Schmidt bi-orthogonal algorithm, making use of the generalized nullspace and Fourier spectral method, we developed a stable, efficient, and accurate eigensolver. The solver is matrix-free, and the most time-consuming matrix-vector product can be accelerated with FFT, achieving almost optimal complexity ($O(\mathtt{DOF}\log(\mathtt{DOF}))$ operations). Furthermore, we rigorously proved its numerical stability and spectral convergence. Extensive numerical results confirm the spectral accuracy and efficiency in different dimensions, and we investigated the excitation spectrum (eigenvalues) and Bogoliubov amplitudes (eigenfunctions) around the ground state with different parameters. Extensions to BdG excitations of other BECs, such as multi-component BEC, spinor-dipolar BEC, etc., are feasible with minor adaptations on the specific matrix-vector product.

Appendix A. Supplement to the convergence analysis. The coerciveness and conformal finite-dimension subspace approximation of the original BdG problem (i.e., (3.16) and (3.18)) are essential properties to establish error estimates; therefore, without loss of generality, we shall consider the following equivalent but much simpler eigenvalue problem: find $(\lambda, u) \in \mathbb{R} \times H_p^m(\Omega)$ such that

$$\mathcal{L}u := -\Delta u + \gamma(\boldsymbol{x})u = \lambda u, \tag{A.1}$$

with constraint $\|u\| = 1$ in one dimension space. Furthermore, we assume the operator $\mathcal{L}$ is symmetric positive semidefinite and $\gamma(\boldsymbol{x})$ is a smooth function. Extensions to high-dimensional (2D/3D) cases are straightforward, and we shall omit them for brevity.

Define the following inner products:

$$a(u,v) = \langle \nabla u, \nabla v\rangle + \langle \gamma u, v\rangle, \quad b(u,v) = \langle u, v\rangle \quad \forall\, u, v \in H_p^m(\Omega),$$
$$\langle u, v\rangle = \int_\Omega u\bar{v}\mathrm{d}x \ \text{ and } \ \langle \nabla u, \nabla v\rangle = \int_\Omega \nabla u \cdot \overline{\nabla v}\mathrm{d}x.$$

It is easy to prove that $a(\cdot,\cdot)$ is an inner product and coercive in $H_p^m(\Omega)$, that is,

$$a(v,v) \le C_a\|v\|_1^2 \ \ \forall\, v \in H_p^m(\Omega),$$

where C_a is a constant. In cases where $\mathrm{null}(\mathcal{L})$ is nonempty, we define $V := \mathrm{null}(\mathcal{L})^\perp = \{v \in H_p^m(\Omega) \,|\, \langle v, w\rangle = 0,\ w \in \mathrm{null}(\mathcal{L})\}$ as the orthogonal complementary subspace, and the coerciveness (V-ellipticity) is satisfied as

$$C_a\|v\|_1^2 \le a(v,v) \ \ \forall\, v \in V. \tag{A.2}$$

In this appendix, we are concerned only with nonzero eigenvalues and their corresponding eigenfunctions, that is, solving the following weak form of (A.1): find $(\lambda, u) \in \mathbb{R} \times V$ such that

$$a(u,v) = \lambda\, b(u,v) \quad \forall\, v \in V, \tag{A.3}$$

subject to constraint $b(u,u) = 1$.

From Lemma 3.3, we could define the approximation finite-dimensional spaces:

$$V_N := X_N \cap V \subset V.$$

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.

Downloaded 07/11/26 to 202.113.11.53 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

The spectral-collocation method for (A.1) amounts to finding $(\lambda_N, u_N) \in \mathbb{R} \times V_N$ such that

$$-\Delta u_N(x_n) + \gamma(x_n)\ u_N(x_n) = \lambda_N\, u_N(x_n), \quad 0 \le n \le N-1, \tag{A.4}$$

with constraints $b(u_N, u_N) = 1$ and $x_n = -L + nh_x \in \mathcal{T}_{\boldsymbol{x}}$. For convergence analysis, it is necessary to consider the following approximation problem: find $(\lambda_N, u_N) \in \mathbb{R} \times V_N$ such that

$$a_N(u_N, v_N) = \lambda_N\, b_N(u_N, v_N) \quad \forall\, v_N \in V_N, \tag{A.5}$$

subject to constraint $b_N(u_N, u_N) = 1$ with

$$\begin{aligned} a_N(u_N, v_N) &:= \langle \nabla u_N, \nabla v_N \rangle_N + \langle \gamma\, u_N, v_N \rangle_N, \quad b_N(u_N, v_N) := \langle u_N, v_N \rangle_N, \\ \langle u, v \rangle_N &:= \frac{2L}{N} \sum_{n=0}^{N-1} u_N(x_n) \overline{v_N}(x_n), \\ \langle \nabla u_N, \nabla v_N \rangle_N &:= \frac{2L}{N} \sum_{n=0}^{N-1} \nabla u_N(x_n) \cdot \overline{\nabla v_N}(x_n). \end{aligned}$$

LEMMA A.1. *The discrete problem* (A.4) *and the discrete variational problem* (A.5) *are equivalent.*

Proof. Multiplying both sides of (A.4) by $\overline{W_j}(x_\ell) = e^{-\mathrm{i}\mu_j(x_\ell + L)}$ and adding up ℓ from 1 to $N-1$, we have

$$\langle -\Delta u_N, W_j \rangle_N + \langle \gamma\, u_N, W_j \rangle_N = \lambda_N \langle u_N, W_j \rangle_N .$$

Setting $u_N(x) = \sum_{j=-N/2}^{N/2-1} c_j W_j(x)$, by the orthogonality of basis $W_j(x)$, we have

$$\langle \nabla u_N, \nabla W_j \rangle_N = \sum\nolimits_{k=-N/2}^{N/2-1} c_k \mu_k \mu_j \langle W_k, W_j \rangle_N = c_j \mu_j^2 = \langle -\Delta u_N, W_j \rangle_N .$$

The above identity holds for the basis of V_N; therefore, the equivalence proof is completed. ∎

Before introducing the convergence results, we define the following notation:

$$\delta_N(u) = \inf_{v \in V_N} \left\{ \|u - v\|_1 + \sup_{w \in V_N} \frac{|a(v, w) - a_N(v, w)|}{\|w\|_1} \right\}.$$

LEMMA A.2. *Assume the $u(x)$ is a smooth and compactly supported function, i.e.,* $\mathrm{supp}\{u\} \subsetneq \Omega$*; then the following estimate holds:*

$$\delta_N(u) \lesssim N^{-(m-\sigma)} \max\{|u|_m, |\gamma u|_m\}, \tag{A.6}$$

where $\sigma = \max\{1, d/2\}$.

Proof. From Lemma 3.3, we have

$$\|u - \mathcal{I}_N u\|_1 \lesssim N^{-(m-1)} |u|_m. \tag{A.7}$$

From the definition of X_N, we can obtain the following two equations:

$$b_N(v_N, w_N) = b(v_N, w_N),\ \langle \nabla v_N, \nabla w_N \rangle_N = \langle \nabla v_N, \nabla w_N \rangle\ \ \forall u_N, v_N \in V_N. \tag{A.8}$$

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.

Downloaded 07/11/26 to 202.113.11.53 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

Based on (A.8), we obtain

$$\big|a(v_N,w_N)-a_N(v_N,w_N)\big|=\big|\langle\gamma v_N,w_N\rangle-\langle\gamma v_N,w_N\rangle_N\big| \quad \forall\, v_N,w_N\in V_N.$$

Hence, combining $\mathcal{I}_N u\in V_N$, the following inequalities hold:

$$\begin{aligned}\big|a(\mathcal{I}_N u,w_N)-a_N(\mathcal{I}_N u,w_N)\big|&=\big|\langle\gamma\mathcal{I}_N u,w_N\rangle-\langle\gamma\mathcal{I}_N u,w_N\rangle_N\big|\\&\le\big|\langle\gamma\mathcal{I}_N u,w_N\rangle-\langle\gamma u,w_N\rangle\big|+\big|\langle\gamma u,w_N\rangle-\langle\gamma u,w_N\rangle_N\big|\\&\quad+\big|\langle\gamma u,w_N\rangle_N-\langle\gamma\mathcal{I}_N u,w_N\rangle_N\big|\\&:=\ I_1+I_2+I_3.\end{aligned}$$

For the first term, combining Lemma 3.3, we have

$$I_1=\big|\langle\gamma\mathcal{I}_N u,w_N\rangle-\langle\gamma u,w_N\rangle\big|\le\|\gamma\|_\infty\|\widetilde{u}-u\|_0\|w_N\|_0\lesssim N^{-m}|u|_m\|w_N\|_1.$$

Since $\mathrm{supp}\{u\}\subsetneq\Omega$, $\mathrm{supp}\{\gamma u\}\subsetneq\Omega$, γu is smooth and periodic on Ω. Setting $w_N(x)=\sum_{k=-N/2}^{N/2-1}c_kW_k(x)$ and combining [30, Theorem 2.3], the following estimates hold:

$$\begin{aligned}I_2&=\big|\langle\gamma u,w_N\rangle-\langle\gamma u,w_N\rangle_N\big|\le\sum_{k=-N/2}^{N/2-1}|c_k|\big|\langle\gamma u,W_k\rangle-\langle\gamma u,W_k\rangle_N\big|\\&\le\sum_{k=-N/2}^{N/2-1}|c_k|\,|(\widehat{\gamma\widetilde{u}})_k-(\widehat{\gamma\widetilde{u}})_k|\le\sqrt{\sum_j|(\widehat{\gamma\widetilde{u}})_j-(\widehat{\gamma\widetilde{u}})_j|^2}\sqrt{\sum_j|w_j|^2}\\&\lesssim N^{-m}|\gamma u|_m\|w_N\|_0\le N^{-m}|\gamma u|_m\|w_N\|_1,\end{aligned}$$

where $(\widehat{\gamma\widetilde{u}})_k=\int_\Omega\gamma(x)u(x)e^{-\mathrm{i}kx}\mathrm{d}x$. For the third term, combining Lemma 3.3, we obtain

$$I_3=\big|\langle\gamma u,w_N\rangle_N-\langle\gamma\mathcal{I}_N u,w_N\rangle_N\big|\le\|\gamma\|_\infty\|u-\widetilde{u}\|_\infty\|w_N\|_0\lesssim N^{-(m-d/2)}|u|_m\|w_N\|_1.$$

Based on the definition of $\delta_N(u)$ and the above error estimates, we have

$$\begin{aligned}\delta_N(u)&\lesssim N^{-(m-1)}|u|_m+N^{-m}|u|_m+N^{-m}|\gamma u|_m+N^{-(m-d/2)}|u|_m\\&\lesssim N^{-(m-\sigma)}\max\{|u|_m,|\gamma u|_m\}.\end{aligned}$$

∎

Based on (A.2) and the general theory of the error estimates for the eigenvalue problems by the approximation method [2, section 8] and [30, section 1], we have the following convergence results.

LEMMA A.3. *For any eigenpair approximation* (λ_N,u_N) *of* (A.4)*, there is an eigenpair* (λ,u) *of* (A.3) *corresponding to* λ *such that*

$$\|u-u_N\|_1\lesssim\delta_N(u),\quad \|u-u_N\|\lesssim\zeta_N\|u-u_N\|_1,\quad |\lambda-\lambda_N|\lesssim\|u-u_N\|_1^2,$$

where $\zeta_N\to0$ *as* $N\to\infty$.

Acknowledgment. We would like to acknowledge Ms. Tongxin Niu for her valuable assistance in performing the numerical comparisons between our algorithm and LOBP4DCG.

REFERENCES

[1] X. ANTOINE, Q. TANG, AND Y. ZHANG, *A preconditioned conjugated gradient method for computing ground states of rotating dipolar Bose-Einstein condensates via kernel truncation method for dipole-dipole interaction evaluation*, Commun. Comput. Phys., 24 (2018), pp. 966–988, https://doi.org/10.4208/cicp.2018.hh80.11.

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.

[2] I. Babuška and J. E. Osborn, *Eigenvalue Problems*, Handbook of Numerical Analysis, Vol. II: Finite Element Methods, North-Holland, Amsterdam, 1991.

[3] Z. Bai and R.-C. Li., *Minimization principles for the linear response eigenvalue problem* I: *Theory*, SIAM J. Matrix Anal. Appl., 33 (2012), pp. 1075–1100, https://doi.org/10.1137/110838960.

[4] Z. Bai and R. C. Li, *Minimization principles for the linear response eigenvalue problem* II: *Computation*, SIAM J. Matrix Anal. Appl., 34 (2013), pp. 392–416, https://doi.org/10.1137/110838972.

[5] D. Baillie, R. M. Wilson, and P. B. Blakie, *Collective excitations of self-bound droplets of a dipolar quantum fluid*, Phys. Rev. Lett., 119 (2017), 255302, https://doi.org/10.1103/PhysRevLett.119.255302.

[6] W. Bao and Y. Cai, *Mathematical models and numerical methods for spinor Bose-Einstein condensates*, Commun. Comput. Phys., 24 (2018), pp. 899–965, https://doi.org/10.4208/cicp.2018.hh80.14.

[7] W. Bao and H. Wang, *A mass and magnetization conservative and energy-diminishing numerical method for computing ground state of spin-*1 *Bose-Einstein condensates*, SIAM J. Numer. Anal., 45 (2007), pp. 2177–2200, https://doi.org/10.1137/070681624.

[8] I. Bloch, J. Dalibard, and W. Zwerger, *Many-body physics with ultracold gases*, Rev. Modern Phys., 80 (2008), pp. 885–964, https://doi.org/10.1103/RevModPhys.80.885.

[9] E. Cancés, R. Chakir, and Y. Maday, *Numerical analysis of nonlinear eigenvalue problems*, J. Sci. Comput., 45 (2010), pp. 90–117, https://doi.org/10.1007/s10915-010-9358-1.

[10] L. Chen, H. Pu, Z.-Q. Yu, and Y. Zhang, *Collective excitation of a trapped Bose-Einstein condensate with spin-orbit coupling*, Phys. Rev. A, 95 (2017), 033616, https://doi.org/10.1103/PhysRevA.95.033616.

[11] I. Danaila, M. A. Khamehchi, V. Gokhroo, P. Engels, and P. G. Kevrekidis, *Vector dark-anti dark solitary waves in multicomponent Bose-Einstein condensates*, Phys. Rev. A, 94 (2016), 053617, https://doi.org/10.1103/PhysRevA.94.053617.

[12] S.-X. Deng, T. Shi, and S. Yi, *Spin excitations in dipolar spin-*1 *condensates*, Phys. Rev. A, 102 (2020), 013305, https://doi.org/10.1103/PhysRevA.102.013305.

[13] M. Edwards, P. A. Ruprecht, K. Burnett, R. J. Dodd, and C. W. Clark, *Collective excitations of atomic Bose-Einstein condensates*, Phys. Rev. Lett., 77 (1996), pp. 1671–1674, https://doi.org/10.1103/PhysRevLett.77.1671.

[14] Y. Gao and Y. Cai, *Numerical methods for Bogoliubov-de Gennes excitations of Bose-Einstein condensates*, J. Comput. Phys., 403 (2020), 109058, https://doi.org/10.1016/j.jcp.2019.109058.

[15] D. S. Hall, M. R. Matthews, J. R. Ensher, C. E. Wieman, and E. A. Cornell, *Dynamics of component separation in a binary mixture of Bose-Einstein condensates*, Phys. Rev. Lett., 81 (1998), pp. 1539–1542, https://doi.org/10.1103/PhysRevLett.81.1539.

[16] T.-L. Ho, *Spinor Bose condensates in optical traps*, Phys. Rev. Lett., 81 (1998), pp. 742–745, https://doi.org/10.1103/PhysRevLett.81.742.

[17] B. Hu, G. Huang, and Y. Ma, *Analytical solutions of the Bogoliubov–de Gennes equations for excitations of a trapped Bose-Einstein-condensed gas*, Phys. Rev. A, 69 (2004), 063608, https://doi.org/10.1103/PhysRevA.69.063608.

[18] J. A. M. Huhtamäki and P. Kuopanportti, *Elementary excitations in dipolar spin-*1 *Bose-Einstein condensates*, Phys. Rev. A, 84 (2011), 043638, https://doi.org/10.1103/PhysRevA.84.043638.

[19] L. Jia, A.-B. Wang, and S. Yi, *Low-lying excitations of vortex lattices in condensates with anisotropic dipole-dipole interaction*, Phys. Rev. A, 97 (2018), 043614, https://doi.org/10.1103/PhysRevA.97.043614.

[20] D. S. Jin, J. R. Ensher, M. R. Matthews, C. E. Wieman, and E. A. Cornell, *Collective excitations of a Bose-Einstein condensate in a dilute gas*, Phys. Rev. Lett., 77 (1996), pp. 420–423, https://doi.org/10.1103/PhysRevLett.77.420.

[21] Y. Kawaguchi and M. Ueda, *Spinor Bose-Einstein condensates*, Phys. Rep., 520 (2012), pp. 253–381, https://doi.org/10.1016/j.physrep.2012.07.005.

[22] R. B. Lehoucq, D. C. Sorensen, and C. Yang, *ARPACK Users' Guide: Solution of Large-Scale Eigenvalue Problems with Implicitly Restarted Arnoldi Methods*, SIAM, Philadelphia, 1998, https://doi.org/10.1137/1.9780898719628.

[23] Y. Li, Z. Wang, and Y. Zhang, *A Bi-orthogonal Structure-Preserving Eigensolver for Large-Scale Linear Response Eigenvalue Problem*, preprint, arXiv:2506.08355, 2025.

[24] W. Liu and Y. Cai, *Normalized gradient flow with Lagrange multiplier for computing ground states of Bose–Einstein condensates*, SIAM J. Sci. Comput., 43 (2021), pp. B219–B242, https://doi.org/10.1137/20M1328002.

Downloaded 07/11/26 to 202.113.11.53 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.

[25] X. Liu, Q. Tang, S. Zhang, and Y. Zhang, *On optimal zero-padding of kernel truncation method*, SIAM J. Sci. Comput., 46 (2024), pp. 23–49, https://doi.org/10.1137/23M1550803.

[26] M. R. Matthews, D. S. Hall, D. S. Jin, J. R. Ensher, C. E. Wieman, E. A. Cornell, F. Dalfovo, C. Minniti, and S. Stringari, *Dynamical response of a Bose-Einstein condensate to a discontinuous change in internal state*, Phys. Rev. Lett., 81 (1998), pp. 243–247, https://doi.org/10.1103/PhysRevLett.81.243.

[27] T. Ohmi and K. Machida, *Bose-Einstein condensation with internal degrees of freedom in alkali atom gases*, J. Phys. Soc. Jpn., 67 (1998), pp. 1822–1825, https://doi.org/10.1143/JPSJ.67.1822.

[28] G. Sadaka, V. Kalt, I. Danaila, and F. Hecht, *A finite element toolbox for the Bogoliubov-de Gennes stability analysis of Bose-Einstein condensates*, Comput. Phys. Commun., 294 (2024), 108948, https://doi.org/10.1016/j.cpc.2023.108948.

[29] M. Shao, F. H. da Jornada, L. Lin, C. Yang, J. Deslippe, and S. G. Louie, *A structure preserving Lanczos algorithm for computing the optical absorption spectrum*, SIAM J. Matrix Anal. Appl., 39 (2018), pp. 683–711, https://doi.org/10.1137/16M1102641.

[30] J. Shen, T. Tang, and L.-L. Wang, *Spectral Methods: Algorithms, Analysis and Applications*, Springer, Berlin, 2011.

[31] D. M. Stamper-Kurn, M. R. Andrews, A. P. Chikkatur, S. Inouye, H.-J. Miesner, J. Stenger, and W. Ketterle, *Optical confinement of a Bose-Einstein condensate*, Phys. Rev. Lett., 80 (1998), pp. 2027–2030, https://doi.org/10.1103/PhysRevLett.80.2027.

[32] D. M. Stamper-Kurn and M. Ueda, *Spinor Bose gases: Symmetries, magnetism and quantum dynamics*, Rev. Modern Phys., 85 (2013), pp. 1191–1244, https://doi.org/10.1103/RevModPhys.85.1191.

[33] J. Stenger, S. Inouye, D. M. Stamper-Kurn, H.-J. Miesner, A. P. Chikkatur, and W. Ketterle, *Spin domains in ground-state Bose-Einstein condensates*, Nature, 396 (1998), pp. 345–348, https://doi.org/10.1038/24567.

[34] S. Stringari, *Collective excitations of a trapped Bose-condensed gas*, Phys. Rev. Lett., 77 (1996), pp. 2360–2363, https://doi.org/10.1103/PhysRevLett.77.2360.

[35] Q. Tang, M. Xie, Y. Zhang, and Y. Zhang, *A spectrally accurate numerical method for computing the Bogoliubov–de Gennes excitations of dipolar Bose–Einstein condensates*, SIAM J. Sci. Comput., 44 (2022), pp. B100–B121, https://doi.org/10.1137/21M1401048.

[36] T. Tian, Y. Cai, X. Wu, and Z. Wen, *Ground states of spin-F Bose–Einstein condensates*, SIAM J. Sci. Comput., 42 (2020), pp. B983–B1013, https://doi.org/10.1137/19M1271117.

[37] R. M. Wilson, S. Ronen, and J. L. Bohn, *Stability and excitations of a dipolar Bose-Einstein condensate with a vortex*, Phys. Rev. A, 79 (2009), 013621, https://doi.org/10.1103/PhysRevA.79.013621.

[38] W. Zhang, S. Yi, and L. You, *Mean field ground state of a spin-1 condensate in a magnetic field*, New J. Phys., 5 (2003), 77, https://doi.org/10.1088/1367-2630/5/1/377.

[39] Y. Q. Zhang, X. Liu, and M. Xie, *Efficient and accurate computation of the Bogoliubov-de Gennes excitations for the quasi-2D dipolar Bose-Einstein Condensates*, East Asian J. Appl. Math., 11 (2021), pp. 686–707, https://doi.org/10.4208/eajam.301120.250221.

Downloaded 07/11/26 to 202.113.11.53 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.